

AZURE ARCHITECTURE

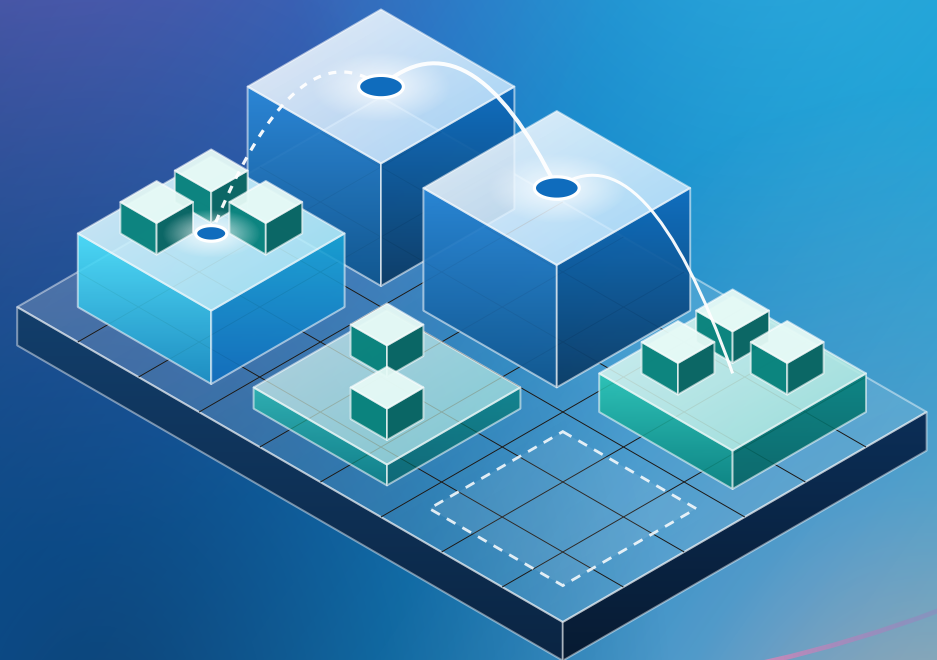
PLATFORM STRATEGY

The adaptable **multi-region** Azure platform

Extending Azure landing zones for adaptable regional placement — qualifying regions, planning by workload archetype, and right-sizing regional platform connectivity.

THREE PLANNING CONSTRUCTS

- Regional Qualification and Enablement Framework
- Application Landing Zone Archetypes
- Regional Platform Connectivity Profiles



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MULTI-REGION PLATFORM WHITEPAPER



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EXECUTIVE SUMMARY

Make regional choice a governed platform capability — not a one-time prediction

Many Azure estates are designed around a fixed primary and disaster-recovery region pair. This paper describes a different approach: extend Azure landing-zone principles so that additional regions become **qualified options that workloads can use selectively**, without creating separate governance models or reproducing an identical platform footprint in every region.

The objective is not to prebuild every capability everywhere. It is to establish a consistent operating model, qualify regional options against known business and workload requirements, and enable only the platform capabilities that each region needs.



Extend, do not replicate

The landing-zone operating model stays consistent across the Azure estate. New regions add only the **most efficient platform capabilities** required by the workloads expected to use them, rather than replicating an existing regional footprint.



Qualify regions against defined requirements

Regional qualification evaluates business and geographic needs, data and compliance constraints, workload requirements and dependencies, regional capabilities, and platform enablement considerations. A region can remain excluded, conditional, or qualified based on documented evidence, constraints, owners, and revalidation triggers.

Qualification creates a **governed regional option**; it does not permanently approve the region for every workload.



Plan by workload requirements, not individual applications

Application Landing Zone archetypes provide a planning abstraction for recurring connectivity and dependency requirements. They help platform teams anticipate what future workloads may require without attempting to predict every application in advance.

Archetypes describe **what applications require**. Regional connectivity profiles describe **what the platform provides**. The relationship is intentionally not one-to-one.

EXECUTIVE SUMMARY • CONTINUED

**Right-size regional connectivity**

Regional Platform Connectivity Profiles range from **Disconnected Spokes** through **Remote Hub Connected** and **Minimal Regional Hub** to a **Full Regional Hub**.

These profiles are not maturity stages. Select the **most efficient profile** for current workload and platform requirements. Add local capabilities only when a measurable need for connectivity, latency, resiliency, security, data handling, or operations justifies them.

Strategic geographic hubs and shared services can continue to support other regions where dependencies and service characteristics permit, avoiding unnecessary duplication.

**Keep platform readiness and workload placement separate**

Regional qualification determines whether a region can become a viable platform option. Actual workload placement occurs later, when a specific workload and its current requirements are known.

During onboarding, validate current Azure service and SKU availability, regional access, quota, dependencies, connectivity, and resiliency requirements. Placement, relocation, recovery, and multi-region operation remain **workload-specific decisions**.

New workload findings can also feed back into regional qualification and trigger platform expansion, additional validation, or reassessment.

**Keep the operating model consistent while allowing regional adaptation**

The operating model stays consistent across regions; implementation can vary where business, workload, or regulatory requirements call for it.

Workload placement, connectivity profiles, local shared-service footprints, Azure services and SKUs, resiliency choices, and other regional parameters can adapt to current requirements.

Region-agnostic does not mean region-random. Regional choice operates within a governed framework of qualified options, documented constraints, and current evidence.

THE GOAL

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**

INTRODUCTION

01

Extending Azure landing zones for adaptable regional placement

The same operating model applies across the estate, while each region is **qualified against the workloads** it may need to support.

Azure landing zones provide a governed foundation for identity, connectivity, security, management, and application delivery. Many organizations initially establish this foundation around a small, fixed set of Azure regions — typically a primary region and a disaster-recovery region, reflecting traditional datacenter and colocation design. This model works while demand and placement requirements remain predictable. It becomes less adaptable as business expansion, data boundaries, service and AI availability, latency requirements, and other regional conditions evolve.

An adaptable multi-region platform extends Azure landing zones without creating separate governance structures or identical infrastructure in every region. The same operating model applies across the estate, while each region is qualified against the workloads it may need to support. Because platform teams cannot predict every future workload, this paper uses three planning constructs: the Regional Qualification and Enablement Framework, Application Landing Zone Archetypes, and Regional Platform Connectivity Profiles.

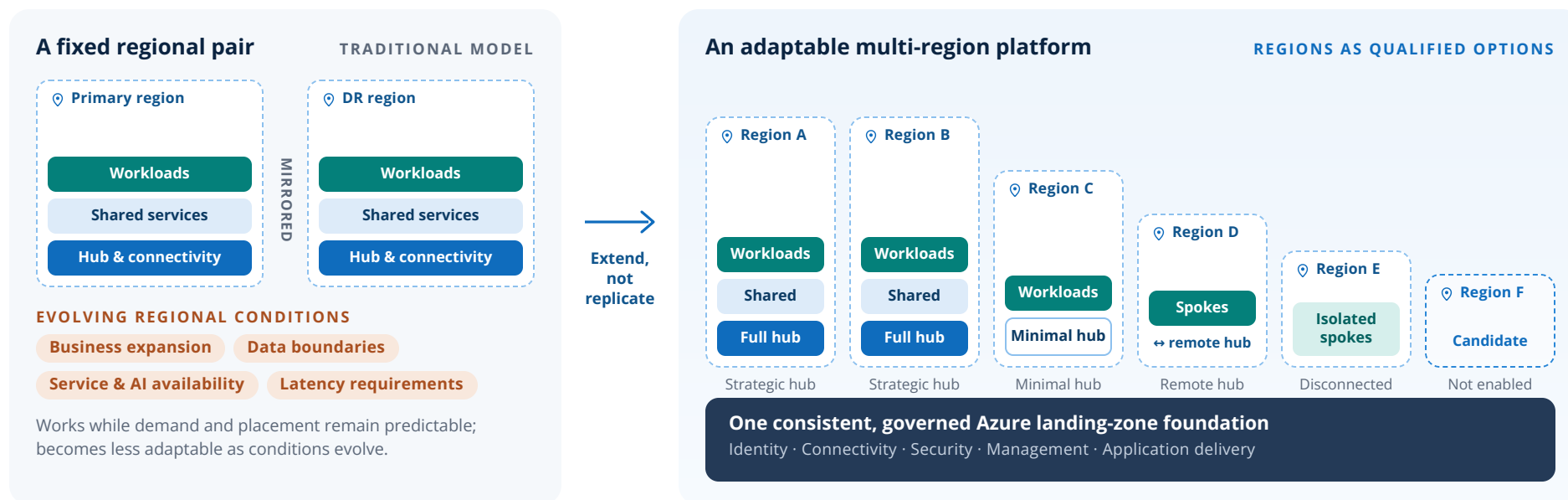


FIGURE 1 From a fixed regional pair to governed regional options. Instead of mirroring a complete footprint in each new region, the platform extends one governed foundation and gives each region only the capability its workloads require. Regional capability is a spectrum, not a copy.

01 · WHAT STARTS THE CONVERSATION

Common drivers of multi-region platform discussions

Multi-region platform work rarely starts as an architecture initiative. It usually starts with a change in the business, regulation, workload requirements, service availability, or risk. These drivers open the discussion; they do not determine the regional architecture or workload-placement outcome.

DRIVER	TYPICAL TRIGGERS
 Business growth and proximity	Expansion into new markets or geographies; acquisitions or divestitures that introduce estates in other regions; digital products that must be closer to customers; user-latency requirements.
 Data residency, sovereignty, and regulation	New or changing data-residency laws; industry requirements for operational resilience or concentration risk; customer contracts; public-sector requirements.
 Service, AI model, SKU, and quota availability	AI models, GPU SKUs, or Azure services available only in selected regions; quota constraints; availability-zone requirements.
 Resiliency and concentration risk	RTO or RPO requirements that the existing regional design cannot satisfy; concentration-risk concerns; paired or nonpaired-region considerations; audit or regulatory findings; lessons from incidents.
 Datacenter and connectivity events	Datacenter or colocation exits; migration waves; office or datacenter moves; network-provider changes; a new Azure region opening within the required geography.
 Cost and operating model	Regional price differences and cross-region transfer cost; landing-zone redesigns or platform refreshes; opportunities to reduce unnecessary regional infrastructure; sustainability objectives.

FIGURE 2 Common drivers of multi-region platform discussions. Typical patterns observed across enterprise estates; each one opens a discussion rather than determining a predetermined outcome.

Drivers open the discussion; they do not determine the regional architecture or the workload-placement outcome.

SCOPE OF THIS PAPER

Azure public-cloud regions. Sovereign cloud offerings and Azure Local have their own constraints and are not covered, although the same qualification logic can inform them.

THREE PLANNING CONSTRUCTS

Balancing advance planning with flexibility

Platform teams cannot predict every future workload. The three constructs let the platform plan for what is known, stay flexible for what is not, and hand workloads a set of governed regional options to choose from.

PLATFORM READINESS · ADVANCE PLANNING

DESCRIBE REQUIREMENTS, QUALIFY REGIONS, DEFINE WHAT THE PLATFORM PROVIDES

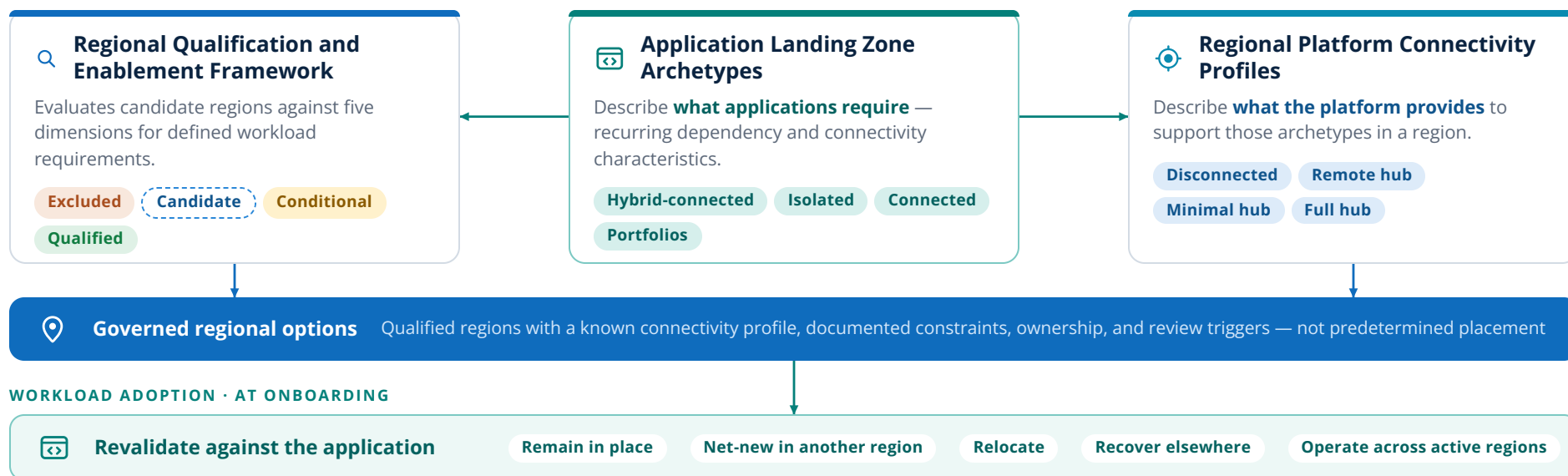


FIGURE 3 How the three constructs connect. Archetypes (application requirements) and connectivity profiles (platform capability) stay distinct; together with regional qualification they produce governed options. Workloads adopt those options only after onboarding revalidation — migration and resiliency remain workload-specific decisions.

REGIONAL QUALIFICATION AND ENABLEMENT FRAMEWORK

Provides a repeatable way to assess candidate regions. It evaluates business and geographic need, data and compliance boundaries, workload requirements and dependencies, regional capabilities, and required platform enablement. The result is an excluded, candidate, conditional, or qualified regional option, backed by current evidence, known constraints, ownership, and review triggers.

APPLICATION LANDING ZONE ARCHETYPES

Group recurring dependency and connectivity characteristics so platform teams can plan before every workload is known. The four archetypes are *Hybrid-Connected Applications*, *Isolated Cloud-Native or AI Applications*, *Connected Cloud-Native or AI Applications*, and *Interconnected Application Portfolios*. Use known requirements for advance planning; defer implementation details that do not affect regional qualification to workload onboarding.

REGIONAL PLATFORM CONNECTIVITY PROFILES

Describe what the platform provides. A region can use Disconnected Spokes, Remote Hub Connected, Minimal Regional Hub, or Full Regional Hub. The most efficient profile can remain the target as long as it continues to satisfy requirements; add local capabilities only when those requirements change. Hybrid connectivity and regional workload connectivity are related but separate decisions, so adopting another region does not by itself require duplicating the existing connectivity platform.

PROGRESSIVE ADOPTION JOURNEY

Establish regional options first — then use and expand them as workload needs justify

Together, these constructs create governed regional options without predetermining workload placement. During onboarding, teams revalidate the region against the workload's current dependencies and requirements, then decide whether to stay put, deploy net-new, relocate, recover elsewhere, or operate across regions.

Migration and resiliency remain workload-specific. The framework makes later regional adoption easier without trying to predict the final estate up front. The journey below shows the recommended sequence for adopting regional options. It is **not a mandatory maturity path**; platform capabilities and workload use evolve only when business and application requirements justify them.

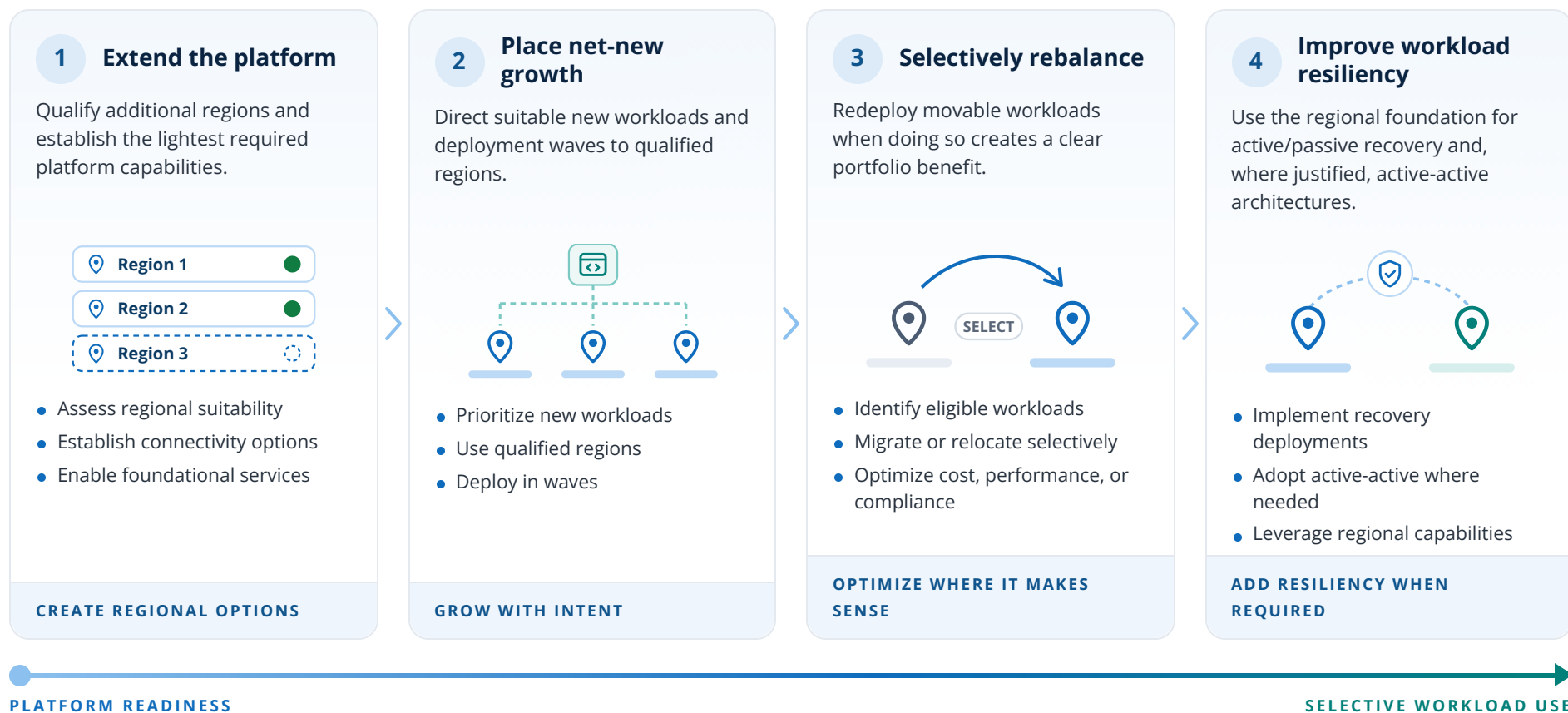


FIGURE 4 A progressive journey for multi-region adoption. Regional options are created first and consumed selectively. Each step is justified by business and application need; a platform can stop at any step that continues to meet requirements.

TWO LAYERS, TWO DECISIONS

02 Multi-region platform and multi-region workload

A multi-region platform and a multi-region workload operate at different layers. The platform creates governed regional choices for the customer estate; the workload applies one or more of those choices to a specific workload. Keeping these concepts separate prevents multi-region strategy from being treated as an automatic requirement for a new landing zone, workload migration, active-active high availability, or active/passive disaster recovery.

The **platform** creates governed regional choices for the estate; the **workload** applies one or more of those choices to a specific application.

PLATFORM READINESS DOES NOT IMPLY

- × A new landing zone
- × Workload migration
- × Active-active HA
- × Active/passive DR

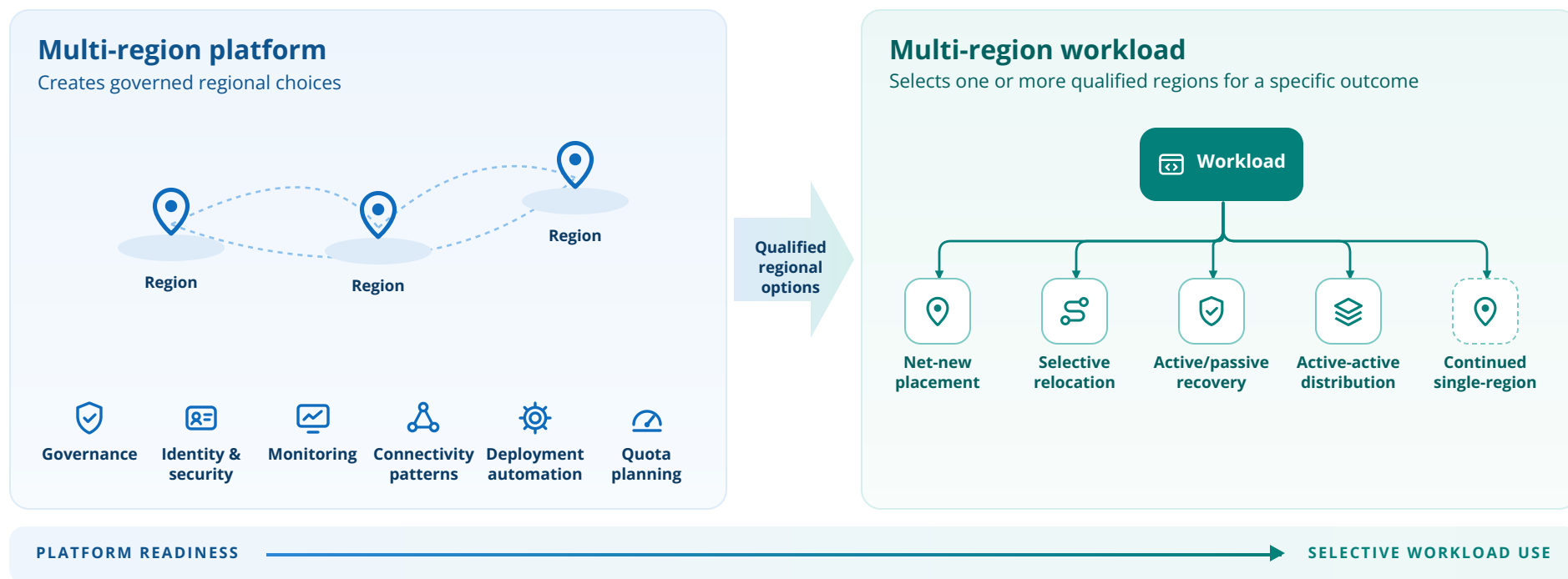


FIGURE 5 Multi-region platform and multi-region workload operate at different layers. Platform capabilities make regions available as qualified options; each workload then selects its own outcome — which can legitimately be continued operation in a single region.

■ A multi-region platform

Qualifies additional regions through the existing landing-zone operating model, approved connectivity patterns, deployment automation, and quota planning. It does not require a separate connectivity landing zone or identical shared services in every region; workload requirements determine the cross-region design.

■ A multi-region workload

Targets the requirements of a specific workload or component. Depending on workload requirements and regional suitability, the outcome might be net-new placement in another region, selective relocation, active/passive recovery, active-active distribution, or continued operation in one region.

Two layers compared

TABLE 1

	MULTI-REGION PLATFORM	MULTI-REGION WORKLOAD
Purpose	Enable placement flexibility across qualified Azure regions	Meet workload-specific objectives and requirements such as availability, disaster recovery, user proximity, or regulatory requirements
Focus	Regional readiness and consistency of shared platform capabilities	Application architecture, data, dependencies, traffic, availability, and recovery
Design driven by	The needs of the portfolio of workloads the platform is expected to support	The specific workload’s requirements, such as availability, resiliency, RTO/RPO, latency, or data requirements

DEFAULT POSTURE

Reuse the existing Azure landing-zone operating model by default. Extend regional policy parameters, connectivity, shared services, automation, and operations only where documented workload requirements justify them.

Multi-region platform readiness establishes the shared capabilities needed to support workloads across selected Azure regions.

Regional suitability depends on service and SKU fit, dependencies, quota, latency, cost, and data constraints; **actual capacity remains a deployment-time condition.**

MULTI-REGION PLATFORM

03 Regional Qualification Framework

Which Azure regions should be **prepared** to support workload requirements — not whether an individual workload should use multiple regions.

This framework determines which Azure regions should be prepared to support workload requirements. It does not determine whether an individual workload should use multiple regions or define its multi-region architecture. Those are workload-specific design decisions. Use the following five dimensions to evaluate candidate regions consistently.

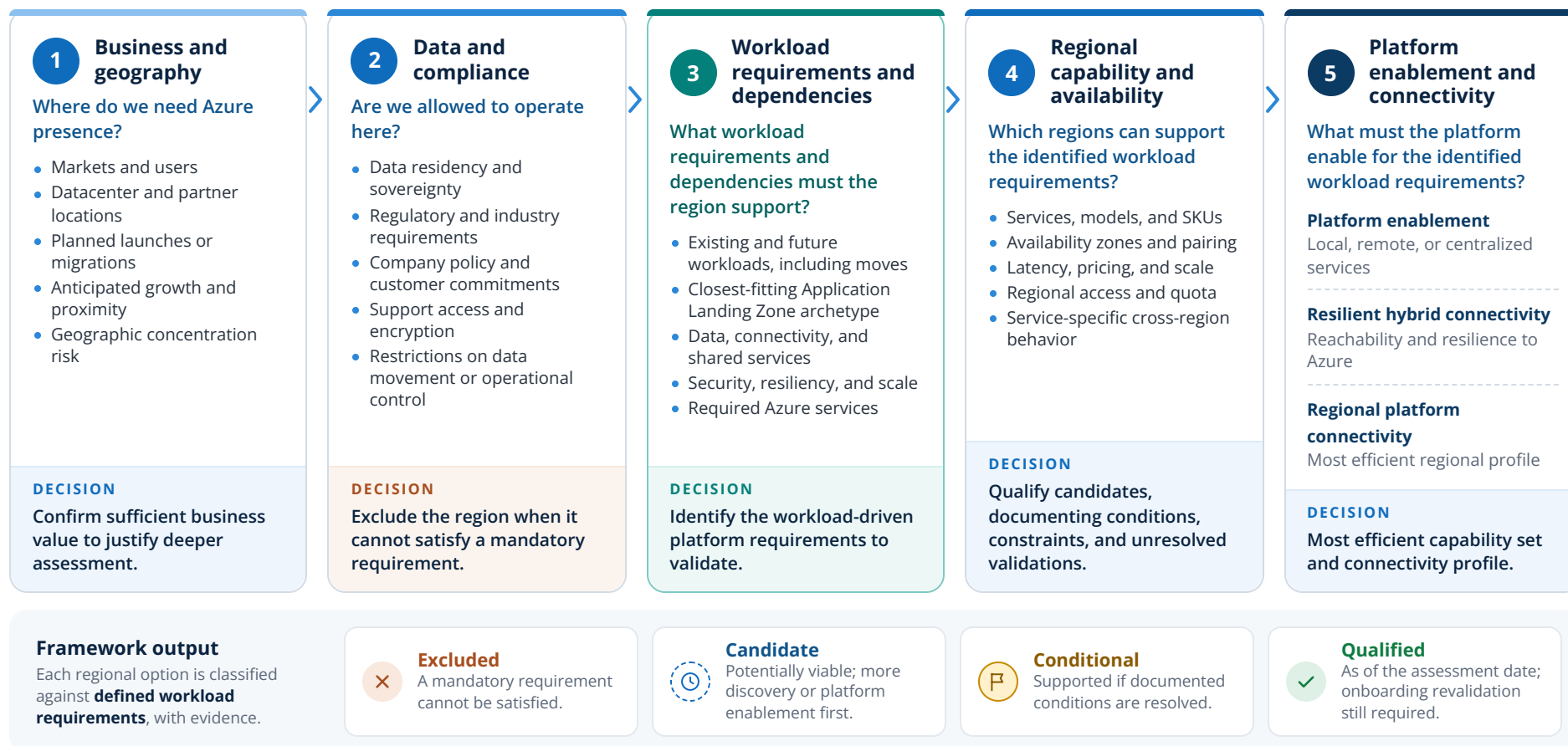


FIGURE 6 Five dimensions — the recommended platform-adoption sequence. Outcomes are specific to defined workload requirements, never a blanket certification; workload resiliency requirements are identified during workload design.

REGIONAL ASSESSMENT OUTCOME

Classify every regional option, with evidence

Consolidate the workload requirements and platform needs identified through the five dimensions. Assess each candidate region against the same criteria and document the result, key tradeoffs, and outstanding validation. Reuse the assessment in architecture reviews and regional planning.

Classify each regional option as:

**Excluded**
A mandatory requirement cannot be satisfied.

**Candidate**
The region is potentially viable, but additional discovery or platform enablement is required before qualification.

**Conditional**
Current evidence supports the defined workload requirements if one or more documented conditions are resolved.

**Qualified as of the assessment date**
Current evidence supports the defined workload requirements, platform ownership is established, and workload-onboarding revalidation is still required.

Record the evidence date, review date, workload requirements in scope, required connectivity profile, key constraints, outstanding validations, owner, and revalidation triggers.

A REPEATABLE INPUT, NOT A ONE-TIME EXERCISE



 Workload onboarding revalidates options and feeds new findings back into the assessment.

Illustrative: one portfolio, classified per archetype

FIGURE 7

CANDIDATE REGION	HYBRID-CONNECTED	ISOLATED CLOUD-NATIVE / AI	CONNECTED CLOUD-NATIVE / AI	INTERCONNECTED PORTFOLIO
Region A strategic hub region				
Region B strategic hub region				
Region C in-country, new				
Region D AI model and SKU				
Region E other geography				
 Qualified  Conditional  Candidate  Excluded				

NOTE The same region can be qualified for one set of requirements, conditional or a candidate for another, and excluded for a third. Qualification is a dated planning input — not permanent approval; decision-time validation at onboarding remains authoritative. Regions and statuses are illustrative.

The purpose is not to certify a region for every possible workload, but to know which options are excluded, which remain candidates or conditional, and which are ready for defined workload requirements.

1 Business and geography

2 Data and compliance

3 Workload requirements

4 Regional capability

5 Platform and connectivity

1 Business and geography

Where do we need Azure presence?

Start with the business reason for considering another region. Evaluate expected markets and users, datacenter and partner locations, planned launches or migrations, anticipated growth, proximity requirements, and geographic concentration risk.

DECISION

Confirm that the region has sufficient business value to justify deeper assessment.

REFERENCE LINKS

- › [Select Azure regions](#)
- › [Azure geographies](#)

2 Data and compliance

Are we allowed to operate here?

Determine whether anticipated workloads and their data can operate in the region. Consider data residency and sovereignty, regulatory and industry requirements, company policy, customer commitments, support access, encryption requirements, and restrictions on data movement or operational control.

DECISION

Exclude a region when it cannot satisfy a mandatory data, compliance, or policy requirement.

REFERENCE LINKS

- › [Data residency in Azure](#)
- › [Azure compliance documentation](#)
- › [Reliability and sovereignty in Azure](#)

CANDIDATE REGIONS UNDER CONSIDERATION

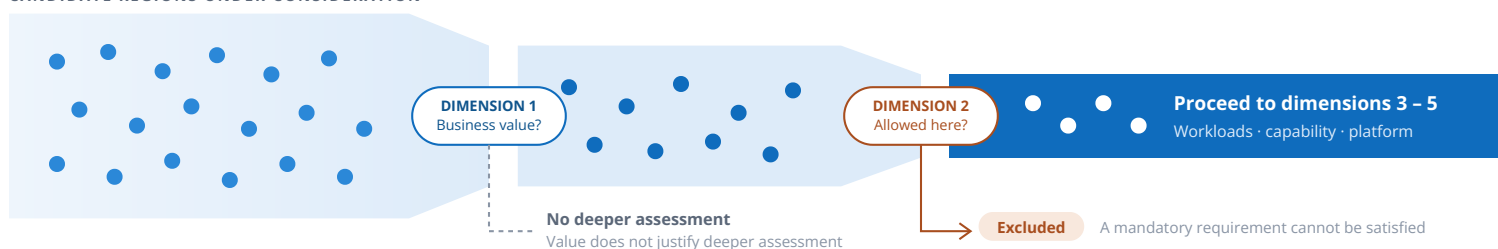


FIGURE 8 The first two dimensions filter the portfolio. Business value decides whether a region merits deeper assessment; mandatory data, compliance, and policy requirements decide whether it can be used at all. Only regions that pass both proceed to workload, capability, and platform analysis.

1 Business and geography

2 Data and compliance

3 Workload requirements

4 Regional capability

5 Platform and connectivity

3 Workload requirements and dependencies

What workload requirements and dependencies must the region support?

Regional qualification should reflect the requirements of workloads expected to consume the region — from existing applications, net-new workload plans, or workloads being considered for relocation between Azure regions.

Review architecture, connectivity, data, shared-service, security, resiliency, scale, and operational requirements and dependencies. Where useful, map each workload to the closest-fitting **Application Landing Zone archetype**; if it has requirements beyond that archetype, record them as explicit additions to it rather than forcing multiple classifications.

For **existing workloads** — in Azure, on-premises, or another cloud — start with the application inventory and architecture. Validate important dependencies with workload owners, first for critical or representative applications and portfolios. Capture application and data dependencies, connectivity, shared services, required Azure services, security and resiliency requirements, and expected scale.

If it would move into the region, from another Azure region or from outside Azure, also identify what must move together, what can stay where it is, and how it behaves while split across locations — latency, data movement, application changes, acceptable disruption, and temporary operation.

For **future workloads**, use known business and technical requirements, predefined workload patterns, intended architecture, expected dependencies, and required services. Where design is still evolving, identify what must influence qualification and defer implementation-specific decisions to onboarding.

DECISION

Identify the workload-driven platform requirements that must be validated as part of regional qualification.

SCOPE BOUNDARY

This phase identifies the regional and platform capabilities workloads require. It does not approve relocation or define migration execution; relocation readiness, sequencing, recovery design, and multi-region operation remain workload-specific.

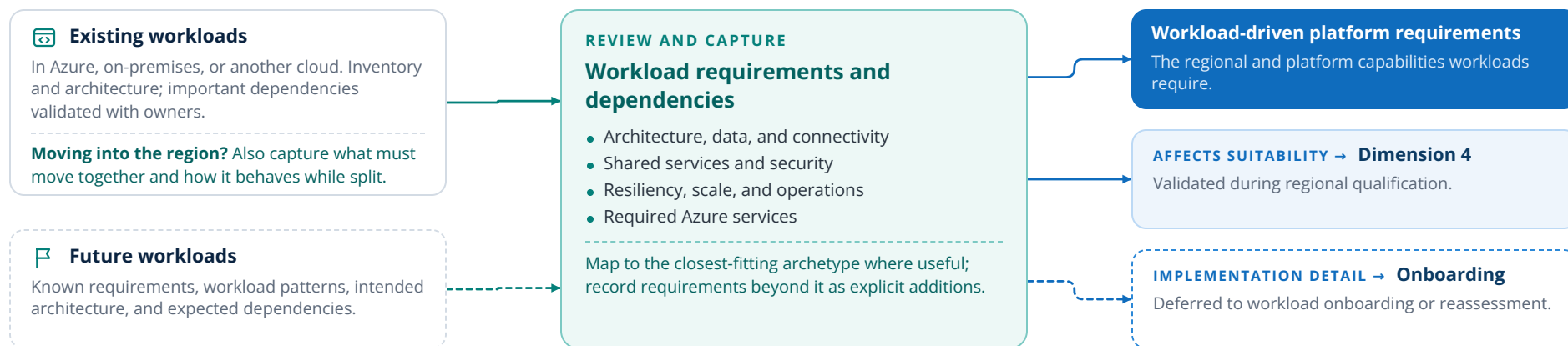


FIGURE 9 From workloads to platform requirements. Requirements that could materially affect regional suitability are validated in Dimension 4; implementation-specific decisions can wait for onboarding.

DIMENSION 3 · ORGANIZING WORKLOAD REQUIREMENTS

Four architecture concepts — related, but not interchangeable

Architecture concepts can help identify and organize workload requirements, but they describe different aspects of a workload and should not be used interchangeably.

TERM	WHAT IT DESCRIBES	EXAMPLES	LEVEL
Architecture style	The fundamental structural organization of an application	Microservices, N-tier, event-driven, web-queue-worker, big compute	Application architecture
Architecture / design pattern	A reusable solution to a recurring technical problem	Circuit Breaker, Retry, CQRS, Strangler Fig, Competing Consumers	Design technique
Workload pattern	A recurring workload behavior or set of operational characteristics that can influence platform requirements	Latency-sensitive, data-intensive, batch-oriented, globally distributed, hybrid-dependent	Workload characteristics
Application Landing Zone archetype	A classification of a workload based on the platform connectivity and dependency capabilities it requires	Hybrid-Connected, Connected Cloud-Native or AI, Isolated Cloud-Native or AI, Interconnected Application Portfolios	Platform-consumption model

FIGURE 10 Architecture concepts used to organize workload requirements. Only the Application Landing Zone archetype describes what a workload needs from the platform; it is the concept used in regional qualification.

NOT THE BUILT-IN LANDING-ZONE ARCHETYPES

The Azure landing zones reference architecture also uses *archetypes*: built-in definitions of what must be true for a landing zone to meet environment and compliance requirements at a given scope. Application Landing Zone archetypes in this paper describe what a workload requires from the platform — a different concept.

ONE ARCHETYPE PER WORKLOAD

Classify each workload by its closest-fitting archetype. Record requirements beyond it as explicit additions to that archetype, rather than forcing the workload into multiple classifications.

REFERENCE LINKS

- › Architecture styles
- › Built-in Azure landing zone archetypes
- › Evaluate a cloud workload for relocation
- › Relocate cloud workloads

1 Business and geography

2 Data and compliance

3 Workload requirements

4 Regional capability

5 Platform and connectivity

4 Regional capability and availability

Which regions can support the identified workload requirements?

Evaluate each candidate region against the workload requirements identified in the previous phase. Consider required Azure services, deployment models and SKUs, availability-zone support, regional service dependencies, latency, pricing, expected scale, regional access, and quota requirements.

Assess **availability-zone support** and **region pairing** separately. Prefer zone-enabled regions when zone resiliency is required, but do not assume that region pairing alone provides workload recovery or that a nonpaired region cannot be used as a recovery destination. Validate service-specific capabilities such as replication, backup and restore, and other regional dependencies against the resiliency requirements identified for the workload.

For requirements that depend on multiple regions, validate the intended regional combination at the service level. This includes built-in paired-region behavior such as Storage GRS and region-of-choice capabilities such as Azure SQL replication. Record material gaps, required changes, and owners for unresolved validation; use the resiliency guidance (Appendix A) for detailed design.

CONSIDERATION A

Availability-zone support

Zone 1

Zone 2

Zone 3

Prefer zone-enabled regions where they meet workload requirements.

CONSIDERATION B

Region pairing



Consider both paired and nonpaired options.

Assess separately

DO NOT ASSUME

✗ ...that a **paired** region provides workload recovery.

✗ ...that a **nonpaired** region cannot support it.

VALIDATE REGIONAL DEPENDENCIES

Storage GRS

Azure SQL
replicationBackup and
restore

Key Vault

Built-in Azure capabilities support the required outcome for the candidate regional combination

Additional workload design is required — record gaps, changes, and validation owners

FIGURE 11 Zones and pairing are independent questions. Neither guarantees nor rules out workload recovery. Service-level detail is in Appendix A.

Continued on the next page: regional access, quota, programmatic evaluation, and the Dimension 4 decision.

1 Business and geography

2 Data and compliance

3 Workload requirements

4 Regional capability

5 Platform and connectivity

DIMENSION 4 · CONTINUED

Access, quota, and evidence-based evaluation

Validate whether the intended subscription and offer types can deploy in the region, and identify any regional access restrictions or approvals that must be resolved before deployment. Assess required quota based on expected workload scale and identify any increases that may be required.

QUOTA IS NOT CAPACITY

Quota should not be treated as an indication or guarantee of Azure capacity, which remains subject to availability at deployment time.

Evaluate objective regional characteristics programmatically, where possible

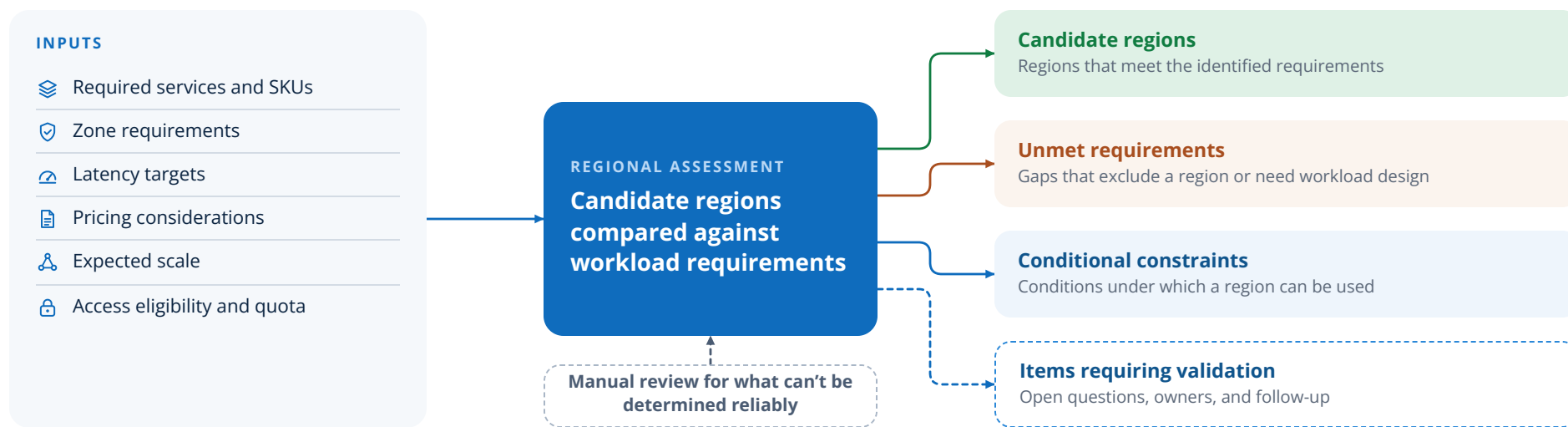


FIGURE 12 Programmatic regional evaluation. Published and programmatically available information narrows the options; manual review covers considerations that cannot be determined reliably that way.

If a requirement that could materially affect regional suitability is unresolved, retain the region as a **conditional candidate** until it is validated. Evaluate requirements that do not affect regional qualification later during workload onboarding or reassessment.

DECISION

Qualify candidate regions against identified workload requirements, documenting conditions, constraints, and unresolved validations while preserving viable options for future workloads.

REFERENCE LINKS

- › [Azure region pairs and nonpaired regions](#)
- › [Azure services that support multiple regions](#)
- › [Azure services that support availability zones](#)
- › [Zonal resources and zone resiliency](#)
- › [Enable zone resiliency for Azure workloads](#)

1 Business and geography

2 Data and compliance

3 Workload requirements

4 Regional capability

5 Platform and connectivity

5 Platform enablement and connectivity

Platform enablement

What platform capabilities must be enabled to support the identified workload requirements?

Determine which shared services and operational capabilities must be provided in the region, consumed from another region, or remain centralized. Consider identity and access, policy and security controls, deployment automation, monitoring and logging, backup and recovery, key and secret management, operational support, platform resiliency, ownership, and cost.

Qualifying a region does not require recreating the full platform footprint used elsewhere. Enable only the regional capabilities required by expected workloads, and document which services remain centralized or are consumed remotely. Define the conditions that would trigger future expansion.

DECISION

Define the most efficient platform capability set, where each capability will be provided, who will operate it, and what requirements would trigger future expansion.

REFERENCE LINK

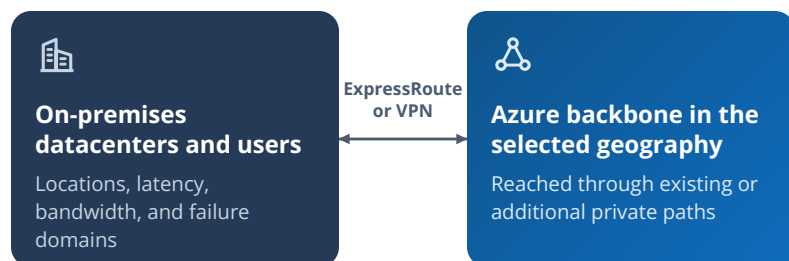
[Azure landing zone design areas and conceptual architecture](#)

Connectivity

Connectivity planning separates three decisions: **resilient hybrid connectivity** to Azure, placement of **geographic hubs and shared services**, and **application-specific connectivity**. They influence one another, but each has different requirements and tradeoffs. Archetypes can inform the assessment, but the design should follow each workload's actual dependencies and requirements.

- For **existing workloads**, particularly those with legacy or hybrid dependencies, use validated application, network, datacenter, and shared-service dependencies to determine whether existing geographic hubs can continue to support them or additional regional capabilities are required.
- For **new workloads**, derive connectivity needs from the intended architecture and known dependencies, using the archetypes as a planning aid. Reuse existing hubs, shared services, and platform capabilities where they meet those needs; a new region does not automatically require a new platform footprint.
- Existing and new workloads follow the same rule: **requirements and dependencies determine the regional connectivity profile**. Validate workload-specific fit during onboarding and add regional capabilities only when justified.

1 · RESILIENT HYBRID CONNECTIVITY



2 · GEOGRAPHIC HUBS AND SHARED SERVICES



3 · APPLICATION-SPECIFIC REQUIREMENTS



FIGURE 13 Three connectivity planning layers — related, but not the same design decision. The next page examines layer 1 and layers 2–3 separately.

LAYER 1

Resilient hybrid connectivity

Does the operating geography have resilient hybrid connectivity to Azure that satisfies the identified workload requirements?

Evaluate existing hybrid connectivity against datacenter and user locations, data boundaries, provider and peering-location diversity, latency, bandwidth, routing, and failure-domain requirements. Enabling an additional Azure region does not inherently require another ExpressRoute circuit. Determine whether existing ExpressRoute, VPN, or other hybrid connectivity provides the required reachability and resilience.

THE DECIDING FACTOR

Are additional circuits, peering locations, gateways, or paths needed for diversity, failure-domain isolation, latency, bandwidth, routing, or resiliency?

No

Existing connectivity can remain sufficient — reuse the current hybrid foundation.

Yes

Change the foundation — reachability, resiliency, diversity, scale, or ownership.

DECISION

Confirm the hybrid connectivity foundation and identify any required changes to reachability, resiliency, diversity, scale, or ownership.

REFERENCE LINK

[Designing for disaster recovery with ExpressRoute private peering](#)

LAYERS 2 - 3

Regional platform connectivity

What is the most efficient regional platform design that supports the identified scenarios?

Determine how workloads in the region will reach geographic hubs, shared platform services, application dependencies, other Azure regions, and hybrid environments. Select the profile based on those connectivity and dependency requirements:



Choose the profile that meets the requirements without unnecessary infrastructure. Adoption growth alone does not justify moving toward a Full Regional Hub; expand local capabilities when shared-service placement, resiliency, scale, connectivity, or operational requirements change. Profiles are described in Section 05.

DECISION

Select the most efficient regional platform connectivity profile that satisfies the identified requirements and can evolve as regional adoption changes.

REFERENCE LINKS

- › [Network topology and connectivity](#)
- › [Define an Azure network topology](#)

HYBRID



REGIONAL

Geographic hybrid connectivity and workload connectivity remain related but distinct decisions. Enabling another region does not automatically require duplicating the existing connectivity platform.

03 · DIMENSION 5 · BEFORE WORKLOADS ARRIVE

Platform readiness checklist for a new region

These are the platform items that most often stall a new region after it has been qualified. Most are inexpensive to plan early and expensive to retrofit. Apply only those the region's connectivity profile and workload requirements call for.



Network foundation

- ☐ **IP address planning**
Reserve non-overlapping address space for future regions and profiles. The most common blocker.
- ☐ **Private DNS across regions**
Resolution for private endpoints and hybrid names, wherever the workload runs.
- ☐ **Routing and inspection paths**
How traffic reaches hubs and firewalls without asymmetric paths.
- ☐ **Hybrid connectivity**
Reuse existing circuits or add paths, per Dimension 5.



Governance and automation

- ☐ **Region settings in policy**
Allowed-locations and region parameters that admit the new region only for its qualified workload requirements.
- ☐ **Region-aware infrastructure as code**
Modules, naming, and tagging parameterized by region — no hard-coded locations.
- ☐ **Landing-zone provisioning**
Subscription or landing-zone vending can place workloads in the region.



Operations and data

- ☐ **Where logs and telemetry live**
Workspace location and residency for monitoring and security data.
- ☐ **Backup placement**
Vaults and valid restore targets for the region — see Appendix A.
- ☐ **Runbooks and support**
Incident, on-call, and support procedures include the region.



Capacity and security

- ☐ **Quota and specific capacity**
Quota for the required SKUs and models requested ahead of onboarding; capacity reservations where justified.
- ☐ **Key Vault placement**
Workload vaults per region rather than cross-region dependency — see Appendix A.
- ☐ **Security coverage**
Security posture, threat detection, and privileged access paths extend to the region.

☐ Most common blockers — plan before qualification completes ☐ Confirm as the capability set is enabled

FIGURE 14 Platform readiness checklist. Hub and shared-service implementations differ between organizations; the checklist names what must be true, not how a specific hub is built.

✓ **Ready for workloads** means the region is qualified for its workload requirements, the required capability set is in place through standard automation, and workloads can onboard without platform exceptions.

DECISION TOOL · REGIONAL QUALIFICATION

Should this region be qualified?

Apply per candidate region and set of defined workload requirements. Each decision point is a framework dimension; each branch ends in one of four classifications.

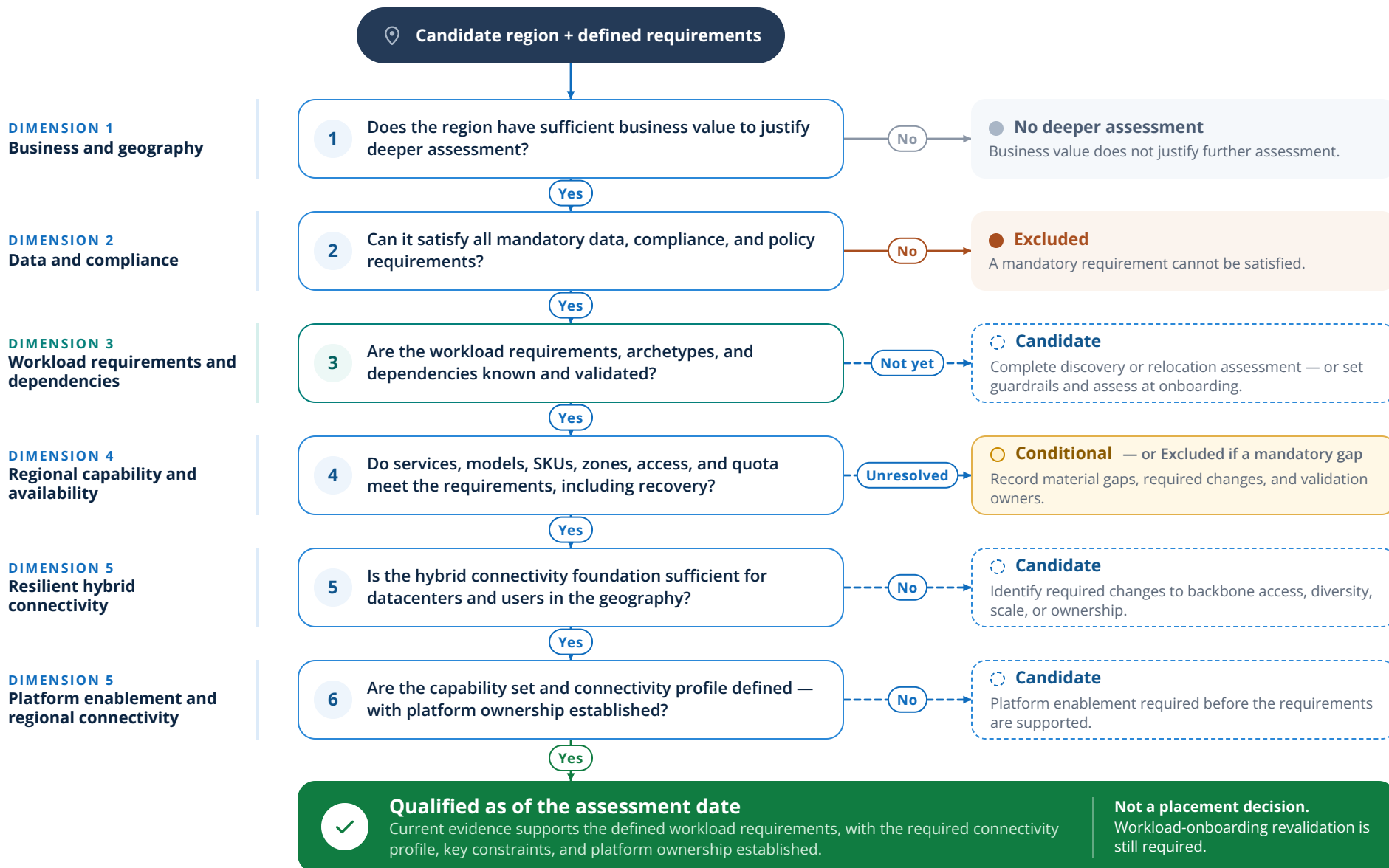


FIGURE 15 Regional qualification decision tree. Filters first, then technical fit, then platform readiness. Candidate and conditional options return to the tree when validation or enablement is complete, or conditions change.

04 WHAT APPLICATIONS REQUIRE

Application Landing Zone Archetypes

A planning abstraction for grouping recurring application connectivity and dependency requirements.

In this paper, an Application Landing Zone archetype describes what an application or application portfolio requires from the platform; it does not replace the existing Azure landing-zone hierarchy or prescribe a specific regional connectivity profile.

Use known workload requirements and application portfolios to anticipate the capabilities that candidate regions may need to support. Where future requirements are not yet sufficiently defined, use the archetypes to establish planning guardrails without unnecessarily constraining future placement decisions.

Archetypes describe what applications require; connectivity profiles describe what the platform provides. The mapping is not one-to-one. For planning, assign the closest-fitting archetype and record material requirements that fall outside it. Portfolio-level dependencies can also affect the platform capabilities needed across several workloads.

Use archetypes during regional qualification to surface recurring requirements that a candidate region may need to support. Placement still happens later, when the workload's current requirements, regional capabilities, and platform readiness are known.

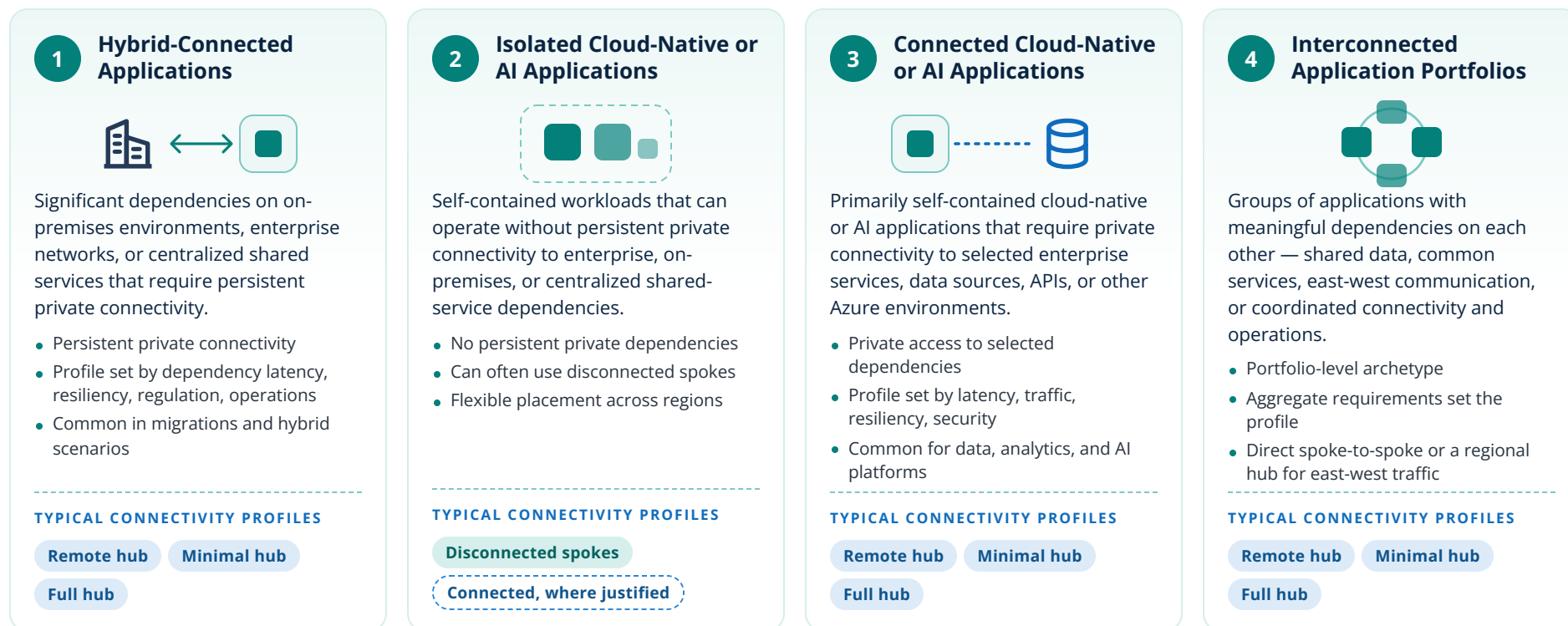


FIGURE 16 Four Application Landing Zone archetypes. Grouped by the platform connectivity and dependency capabilities each requires. Profile relationships are detailed in Figure 17.

ARCHETYPES AND PROFILES

Archetypes describe what applications require. Profiles describe what the platform provides.

The mapping is not one-to-one: one archetype can use different profiles, and one profile can support multiple archetypes. Keeping the two layers separate lets the platform plan capability without prescribing application designs.

ARCHETYPES WHAT APPLICATIONS REQUIRE	REGIONAL PLATFORM CONNECTIVITY PROFILES · WHAT THE PLATFORM PROVIDES				WHAT DRIVES THE CHOICE
	 Disconnected spokes	 Remote-hub-connected spokes	 Minimal regional hub	 Full regional hub	
1 Hybrid-Connected Applications	—	 Typical	 Typical	 Typical	Dependency latency, resiliency, regulatory, operational, and shared-service requirements.
2 Isolated Cloud-Native or AI Applications	 Often	 Where justified	 Where justified	 Where justified	Often disconnected spokes. A connected profile where private management, security, data access, or centralized services justify connectivity.
3 Connected Cloud-Native or AI Applications	—	 Typical	 Typical	 Typical	Private connectivity to selected dependencies; dependency latency, traffic patterns, resiliency, security, and operational requirements decide which profile.
4 Interconnected Application Portfolios	 Spoke-to-spoke	 Where acceptable	 When justified	 When justified	Aggregate portfolio connectivity, shared-service, resiliency, scale, and operational requirements.
 Typical fit  When specific requirements justify it — Not indicated as typical					

FIGURE 17 Archetype-to-profile relationships are many-to-many. Each row is a set of options, not an assignment; the selected profile follows dependencies, latency, traffic, resiliency, and independence requirements.

Interconnected portfolios: east-west traffic

Where significant east-west traffic exists within a region, evaluate whether direct spoke-to-spoke connectivity or a regional hub is a better path than routing through a remote hub. Direct connectivity may suit a few tightly coupled workloads; a Minimal or Full Regional Hub fits broader routing, inspection, resiliency, or operational needs.

Portfolio-level requirements

Because a portfolio's combined dependencies can require capabilities beyond any single application, select its profile from aggregate connectivity, shared-service, resiliency, scale, and operational requirements — not from a single workload.

05

WHAT THE PLATFORM PROVIDES

Regional Platform Connectivity Profiles

Establish the most efficient profile that supports the workload requirements expected to use the region.

Enabling an additional Azure region does not require changing or duplicating the full platform footprint of an existing geographic landing zone. Instead, establish the **most efficient regional platform connectivity profile** that supports the workload requirements expected to consume the region.

Connectivity profiles describe platform capabilities, not workload designs. The mapping to Application Landing Zone archetypes is not one-to-one; the right profile depends on actual dependencies, latency, resiliency, security, data, and operational requirements.

An organization can maintain a small number of strategic geographic hubs that provide resilient hybrid connectivity and common enterprise services. Other Azure regions can use different regional profiles and consume capabilities from those hubs when workload requirements permit.

Regional enablement is a spectrum — from application spokes with no dependency on a geographic hub, through Remote Hub Connected and Minimal Regional Hub profiles, to a Full Regional Hub with greater local capability and independence. These profiles are **not maturity stages**; a profile can remain the appropriate target architecture as long as it continues to satisfy workload and platform requirements.

NO DEPENDENCY ON A GEOGRAPHIC HUB

GREATER LOCAL CAPABILITY AND INDEPENDENCE

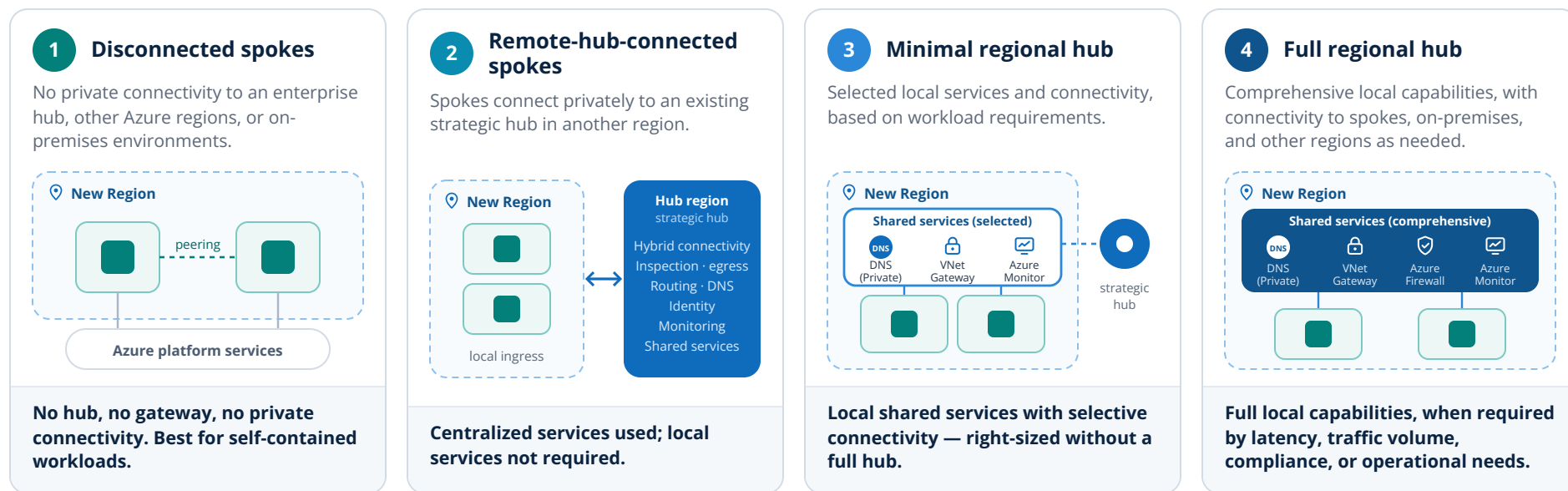


FIGURE 18 The spectrum of regional platform connectivity profiles. Moving right adds local capability and independence — and cost and operational scope. Services shown are examples; each hub includes only what its workloads require. Any profile can be the right long-term target.

1 Disconnected spokes



PROFILE 1 OF 4

Workload spokes operate without private connectivity to an enterprise geographic hub, another Azure region, or an on-premises environment. They can still use direct connectivity to other spokes where required and approved private or public connectivity to Azure platform services.

CHOOSE WHEN

Workloads can operate without private connectivity to an enterprise geographic hub, another Azure region, or an on-premises environment, and their required platform services can be consumed locally or through approved Azure service connectivity. This can include self-contained cloud-native, AI, development, test, batch, HPC, VDI, or internet-facing workloads where those conditions are met.

REGIONAL FOOTPRINT

Workload resources plus the controls needed to deploy, secure, monitor, and operate them. These spokes still use the estate's standard identity, policy, security, and deployment practices.

KEY CONSIDERATIONS

Validate administration, identity, DNS, secrets, telemetry, data movement, software distribution, private endpoints, incident response, and support procedures. Where applications require east-west communication within the region, evaluate direct spoke-to-spoke connectivity without assuming that a regional hub is required. Confirm that no hidden dependency requires connectivity to a geographic hub or on-premises environment.

REASSESS THE PROFILE WHEN

Workloads require private access to enterprise services, centralized security inspection, on-premises systems, another application portfolio, or capabilities hosted through a geographic hub.

2

Remote-hub-connected spokes



PROFILE 2 OF 4

Workload spokes in the additional region use cross-region connectivity to access an existing strategic geographic hub. The geographic hub can provide hybrid connectivity, security inspection, internet egress, routing, DNS, identity, monitoring, or other shared services without requiring a complete hub to be deployed locally. Cross-region connectivity can be implemented through appropriate Azure networking capabilities based on the required topology and routing model. Public application ingress can remain local through services such as Azure Front Door or Application Gateway when appropriate.

CHOOSE WHEN

Workloads require private access to selected enterprise or on-premises services but can tolerate the latency, availability coupling, data-transfer implications, and dependency on connectivity to another region.

REGIONAL FOOTPRINT

Workload spokes and any capabilities that must remain close to the workloads. Gateways, firewalls, hybrid connectivity, routing services, and selected shared services can remain in an existing geographic hub when workload requirements permit.

KEY CONSIDERATIONS

Measure latency and bandwidth using representative traffic. Evaluate cross-region data-transfer cost, routing and transit behavior, inspection paths, geographic hub scale, dependency criticality, and operational ownership. Document how workloads behave if the geographic hub or cross-region connectivity is degraded or unavailable.

REASSESS THE PROFILE WHEN

Cross-region traffic, latency, east-west communication, dependency criticality, regulatory requirements, scale, or availability coupling justify deploying selected connectivity or shared-service capabilities locally.

3 Minimal regional hub



PROFILE 3 OF 4

The additional region includes a limited hub footprint that provides selected local connectivity and shared services while continuing to consume other capabilities from an existing geographic hub. The minimal hub provides only the local capabilities required by its workloads rather than replicating a complete platform footprint. Depending on workload requirements, these can include east-west routing, DNS, monitoring, backup, security inspection, egress, private connectivity, selected identity or directory services, on-premises connectivity, or other services that must remain close to the workloads.

CHOOSE WHEN

Workloads require some local shared services, application-to-application connectivity, lower-latency dependency paths, regional security controls, or greater independence than a remote-hub-connected profile can provide.

REGIONAL FOOTPRINT

A regional hub containing only the connectivity and shared-service capabilities that must operate locally. Hybrid connectivity, specialized networking functions, central management, and other services can remain in a strategic geographic hub when appropriate.

KEY CONSIDERATIONS

Clearly define which capabilities operate locally and which remain remote. Avoid ambiguous routing, overlapping responsibilities, and unnecessary duplication of platform controls. Validate workload and operational behavior when either local capabilities or remote dependencies are unavailable. When on-premises connectivity is required, determine whether existing ExpressRoute circuits and gateway topology can provide the required reachability before introducing additional circuits; **do not assume that each Azure region requires a dedicated ExpressRoute circuit.**

REASSESS THE PROFILE WHEN

Workload growth introduces additional local connectivity or shared-service requirements, multiple interconnected application portfolios require broader regional capabilities, stricter independence requirements emerge, significant hybrid connectivity is required locally, or other requirements justify expanding the regional footprint.

4 Full regional hub



PROFILE 4 OF 4

The region provides the complete set of local connectivity and shared services required for its designated workloads and geographic responsibilities. Depending on those requirements, this can include local hybrid connectivity, security inspection, routing, internet ingress and egress, DNS, identity services, monitoring, backup, key management, and other required platform capabilities.

A full regional hub uses the same operating model and control baselines as other strategic hubs. **It does not have to be an identical copy;** its capabilities should reflect the workloads and geography it supports.

CHOOSE WHEN

Workload and platform requirements justify a comprehensive local capability set — for example, local hybrid connectivity, substantial local shared services, strict latency or data-boundary controls, significant interconnected application portfolios, or sufficient independence to continue operating when another region or geographic hub is unavailable.

REGIONAL FOOTPRINT

The approved set of local connectivity, shared services, operational capabilities, and independent critical paths required by the region's workload portfolio.

KEY CONSIDERATIONS

Confirm that the required level of independence justifies the additional cost and operational scope. Validate automation, observability, support ownership, security operations, configuration consistency, dependency behavior, and lifecycle management. Where on-premises connectivity is required, determine whether existing ExpressRoute circuits can provide the required reachability and resiliency before introducing additional connectivity.

REASSESS THE PROFILE WHEN

Workload demand or requirements decrease, services can be safely centralized, dependencies change, or the full regional footprint no longer provides sufficient value relative to its cost and operational complexity.

05 · PROFILES IN PRACTICE

Cost, independence, and how hubs are built

How the profiles trade local capability for platform scope

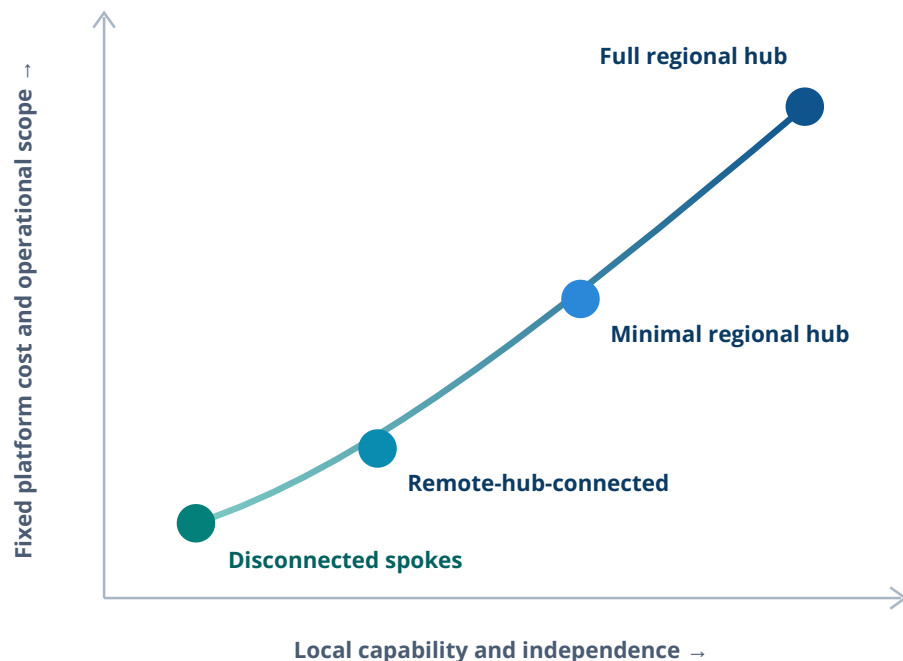


FIGURE 19 A qualitative trade-off, not a price list. Greater local capability and independence usually increase fixed platform cost and operational scope, but they can reduce cross-region dependencies, latency exposure, and availability coupling. There is no required progression between profiles: select the most efficient profile for current requirements.

Regional connectivity profiles describe **capability**, not a specific network build. Two questions matter in practice: what each increase in local capability costs to run, and how the selected hub capabilities are implemented in that organization.

Hub implementations vary between organizations

Azure Firewall hub

Platform-native routing, inspection, and egress with Azure Firewall and related Azure networking.

Workload spokes

Hub · Azure Firewall

NVA firewall hub

Third-party network virtual appliances provide routing, inspection, or security functions.

Workload spokes

Hub · NVA firewalls

SD-WAN-integrated hub

Branch, datacenter, and cloud connectivity integrated through the organization's SD-WAN.

Workload spokes

Hub · SD-WAN + inspection

Separate transit and security functions

Separated where scale, ownership, or architecture requirements justify it.

Workload spokes

Security · inspection

Transit · hybrid, routing

FIGURE 20 The profiles are implementation-neutral. Any of these hub designs can sit behind a strategic, minimal, or full regional hub.

TOPOLOGY NOTES

The profiles can be implemented using hub-and-spoke virtual networks, Azure Virtual WAN, or other approved Azure networking patterns. Topology determines how the required capability is implemented; workload and platform requirements determine which profile is appropriate.

PROFILE SELECTION

Principles for choosing and evolving a profile

- 01 Start with the **workload requirements and dependencies** identified during regional qualification, including application-portfolio dependencies and relevant architecture characteristics.
- 02 Use Application Landing Zone archetypes as a planning abstraction to identify recurring connectivity and shared-service requirements; validate the selected profile against actual workload requirements.
- 03 Select the **most efficient regional connectivity profile** that satisfies the identified requirements.
- 04 Keep capabilities remote when latency, dependency, data, security, resiliency, availability, and operational characteristics permit. Deploy capabilities locally only when a measurable workload or platform requirement justifies the additional footprint.
- 05 Consider direct spoke-to-spoke connectivity where workloads require significant east-west communication and a hub transit path is unnecessary. Introduce regional routing, inspection, or shared services when broader requirements justify them.
- 06 Allow different profiles to coexist across the estate. Profiles evolve only when workload and platform requirements change, not because a region is expected to progress through a predefined maturity path.

A connectivity profile defines the platform capabilities available in a region. **Workload placement, profile evolution, and future qualification remain separate decisions.**

Profiles coexisting across one estate

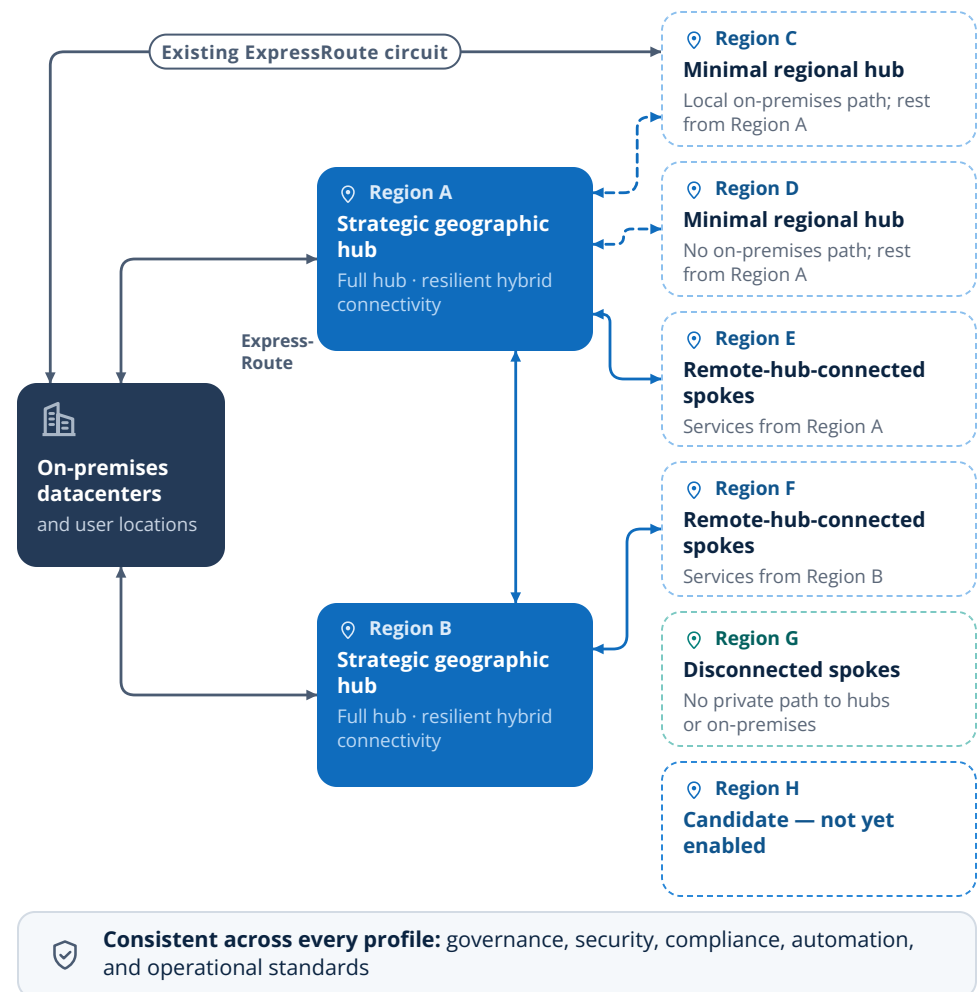


FIGURE 21 An illustrative estate: two strategic hubs provide resilient hybrid connectivity; other regions consume what they need. Minimal hubs take selected services from Region A (dashed) — Region C also reaches on-premises by reusing an existing ExpressRoute circuit, Region D does not. Both are valid. Names and placements are examples only.

DECISION TOOL · CONNECTIVITY PROFILE

Which profile should this region provide?

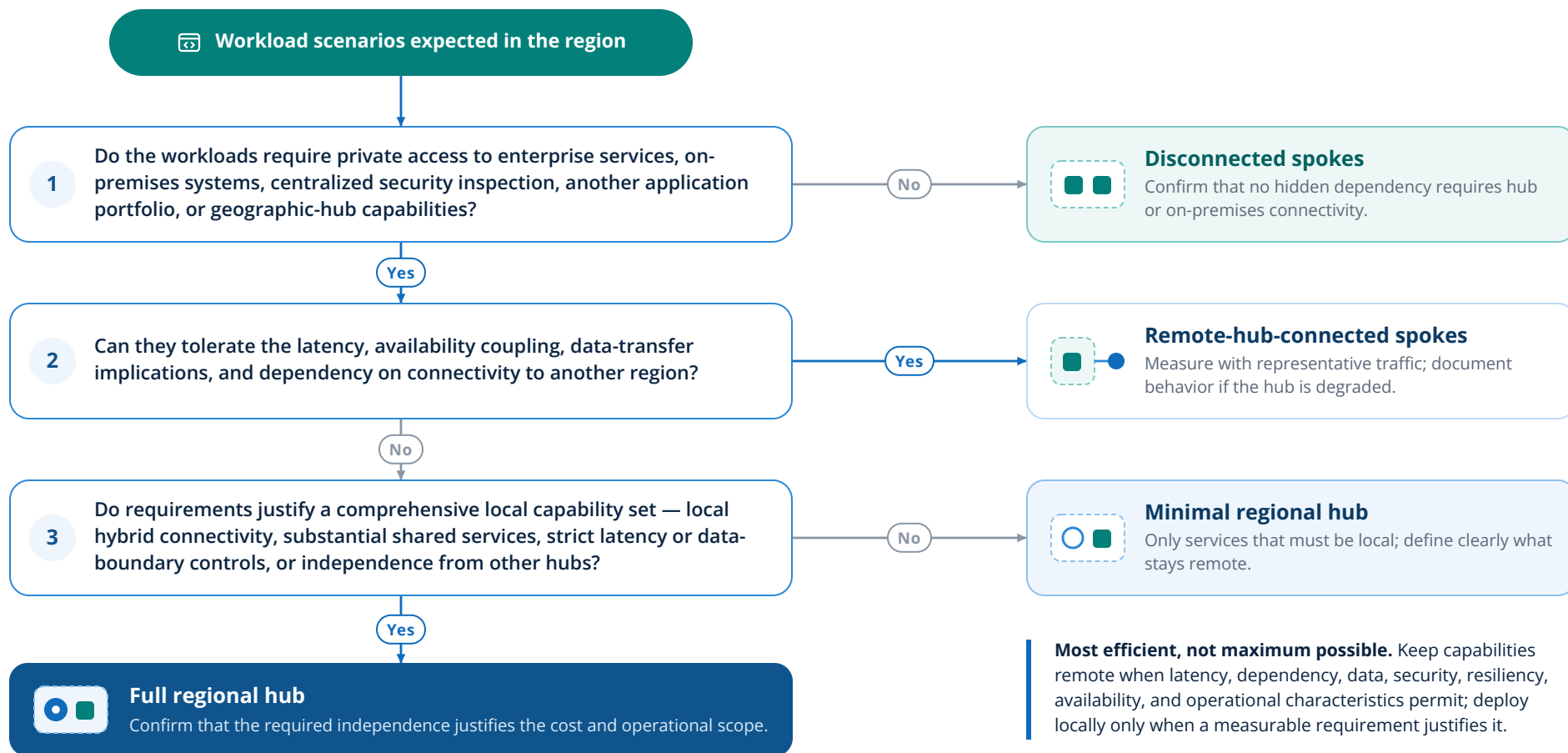


FIGURE 22 Connectivity profile selection tree. Questions escalate from dependency, to tolerance of remote consumption, to required independence. The outcome is a platform capability, not a workload design.

PROFILES EVOLVE IN BOTH DIRECTIONS — REASSESS WHEN

DISCONNECTED SPOKES

Workloads require private access to enterprise services, inspection, on-premises systems, another portfolio, or hub capabilities.

REMOTE-HUB-CONNECTED

Cross-region traffic, latency, east-west communication, dependency criticality, regulation, scale, or availability coupling justify local capabilities.

MINIMAL REGIONAL HUB

Growth adds local requirements, interconnected portfolios need broader capabilities, stricter independence, or significant local hybrid connectivity.

FULL REGIONAL HUB

Demand or requirements decrease, services can be safely centralized, dependencies change, or the footprint no longer justifies its cost.

FROM OPTIONS TO PLACEMENT

06 Workload placement using qualified options

Previous regional qualification is a planning input; **decision-time validation** is authoritative.

Place a workload only after its requirements are understood well enough to make a decision. Regional qualification narrows the choices; it **does not permanently approve** a region for every application. Placement ends in one of these outcomes: place the workload in a qualified region, keep it where it is, relocate it, extend it to another region for recovery or active use, or conclude that no qualified option meets its requirements yet.

Apply the same eight steps to each workload or related application portfolio. The workload team works within the qualified options; additional platform capabilities, where needed, are a separate governed decision. Findings flow back into regional qualification.

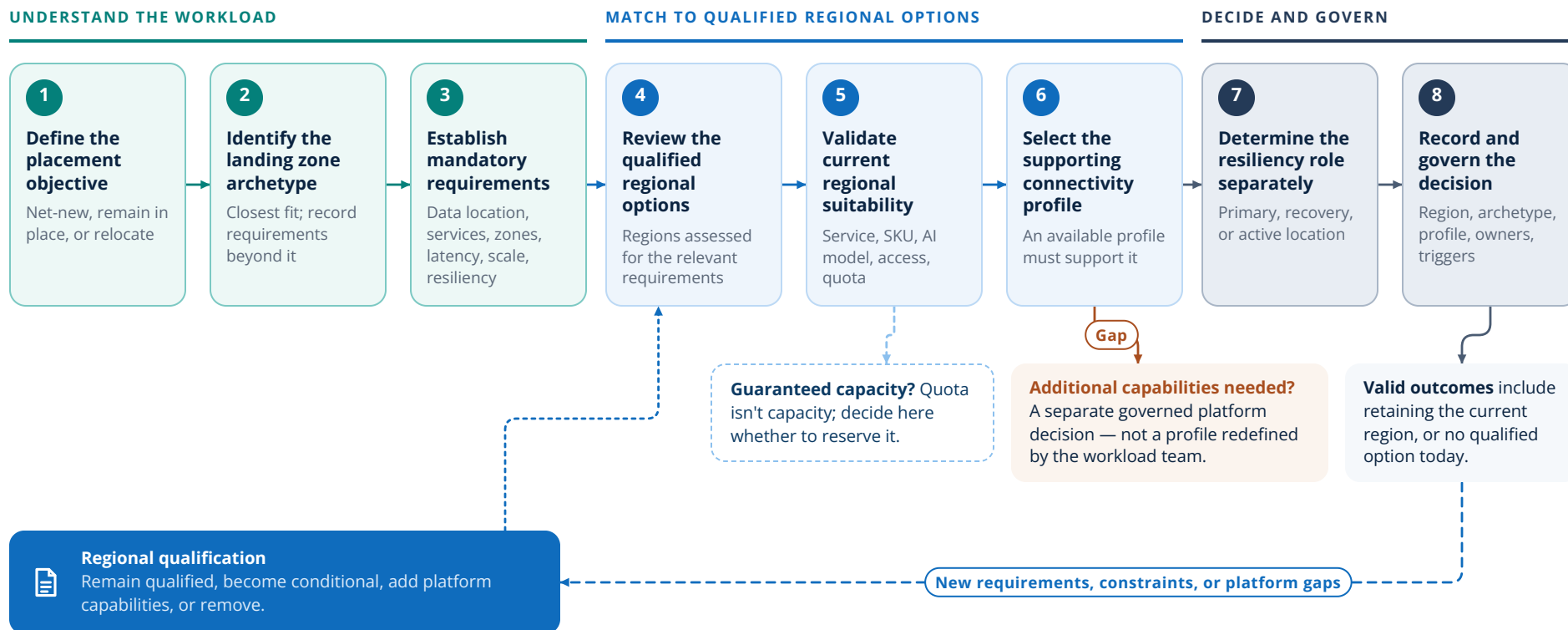


FIGURE 23 Workload placement using qualified options. Teal steps describe the workload, blue steps match it to regional options, and slate steps decide and govern. Onboarding findings return to regional qualification, closing the loop between platform readiness and workload adoption.

THE EIGHT PLACEMENT STEPS

01 Define the placement objective

Is the workload new, staying where it is, or moving or expanding to another region?

02 Identify the Application Landing Zone archetype

Assess the workload's connectivity, shared-service, data, security, operational, and dependency characteristics and pick the closest-fitting archetype. Record any material requirements it doesn't cover as explicit additions to it.

03 Establish mandatory requirements

Confirm data-location restrictions, required Azure services and features, availability-zone requirements, latency thresholds, expected scale, connectivity, resiliency requirements, and critical dependencies.

04 Review the qualified regional options

Consider regions previously assessed against the relevant workload requirements. Review their supported capabilities, connectivity profiles, known constraints, when each was last assessed, and any outstanding validations.

05 Validate current regional suitability

Qualification was an earlier planning assessment, so **re-check the target region before onboarding**, whether the workload is new or moving. Confirm that the services, SKUs, AI models, regional access, quota, and dependencies it needs are available now. Quota allows deployment but doesn't guarantee capacity; if the workload needs guaranteed capacity, decide here whether to reserve it.

06 Select the supporting connectivity profile

Confirm that an available regional connectivity profile satisfies the workload's requirements. If additional platform capabilities are required, treat their introduction as a separate governed platform decision rather than allowing the workload team to redefine the regional profile independently.

07 Determine the resiliency role separately

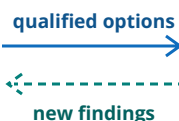
Define whether the region supports the workload as its primary location, recovery location, or part of a multi-region active deployment. Apply workload-specific availability, RTO, RPO, replication, data-consistency, and failover requirements.

08 Record and govern the decision

Document the selected region, archetype, connectivity profile, placement and resiliency outcomes, assumptions, constraints, owners, and revalidation triggers. Feed new regional requirements or platform gaps back into regional qualification.

PLATFORM · QUALIFICATION**Is this region a viable option for these requirements?**

Specific to defined requirements and dated. Answered in advance where requirements are known, from current evidence.

**WORKLOAD · PLACEMENT****Where, and in what role, does this workload run?**

Decided when the actual workload is sufficiently understood, with decision-time validation.

FIGURE 24 Qualification versus placement. The platform qualifies regions against defined requirements; each workload decides where and how it actually runs. Each informs the other.

Regional adaptability still operates within the established landing-zone guardrails and qualified regional options.

Workload onboarding can reveal new regional requirements, constraints, or platform gaps. Use these findings to update regional qualification and determine whether an option should remain qualified, become conditional, require additional platform capabilities, or be removed from consideration for the affected workload requirements.

PRINCIPLE

Regional adaptability does not mean arbitrary placement.

OPERATING MODEL

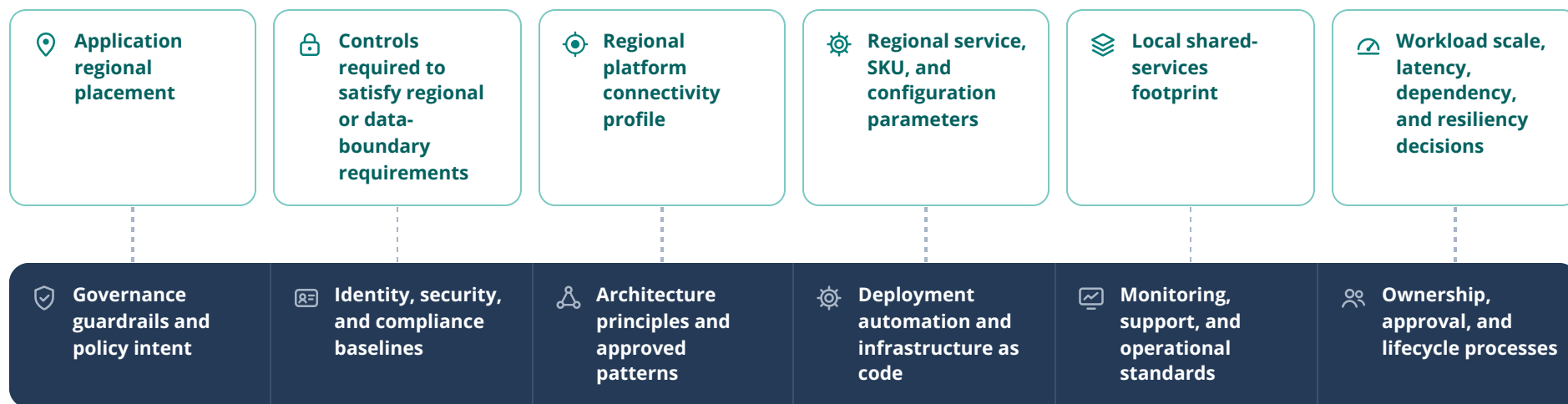
07 A consistent operating model, adaptable by region

Not a different operating model for every region — one model, with governed room to adapt.

A multi-region platform should use one operating model across the estate. Keep governance and architecture standards stable, while regional implementation and workload placement adapt to business, application, and regulatory requirements.

ALLOW TO ADAPT BY REGION AND WORKLOAD

ADAPTS WITHIN THE FOUNDATION'S GUARDRAILS



KEEP CONSISTENT ACROSS THE AZURE ESTATE

ONE FOUNDATION FOR EVERY REGION AND PROFILE

FIGURE 25 What stays consistent and what adapts. Each adaptable decision sits on a consistent foundation: placement varies, governance does not; profiles vary, approved patterns do not.

Region-agnostic does not mean region-random.

Regional adaptability operates within a governed framework. Workloads select from qualified regional options based on their requirements, their Application Landing Zone archetype, and each region's current capabilities and qualification status. Regional qualification should be reassessed as workload requirements and Azure capabilities change.

08

THE FRAMEWORK IN USE

Customer scenarios

How the framework moves from a business or technical driver to a governed regional and workload decision. Illustrative, not customer-specific recommendations.

1

Regulated bank adopts a new in-country region

Residency · nonpaired region

A financial-services organization operates workloads outside its home country. A new Azure region opens in-country while regulatory or business requirements create a need for certain data or services to remain there.

DRIVER

Data residency and regulatory requirements; new in-country Azure region.

**FRAMEWORK**

D2 confirms mandatory location requirements. D3 identifies affected workloads and dependencies. D4 validates required services, zones, access, and constraints.

**QUALIFICATION**

Qualified for defined Hybrid-Connected requirements; other portfolio workloads remain candidates until dependencies are understood.

**ARCHETYPE**

Hybrid-Connected Applications.

**CONNECTIVITY PROFILE**

Minimal Regional Hub where selected local capabilities are required; existing hybrid connectivity reused where reachability and resiliency permit.

**PLACEMENT**

In-scope workloads relocate selectively; others remain in place. Recovery architecture stays workload-specific.

Takeaway. A new in-country region can be qualified for defined workload requirements without forcing portfolio-wide relocation. A nonpaired recovery path must be designed explicitly (Appendix A).

2

AI team requires a specific model or GPU SKU

Specific capacity · net-new

A product team requires an AI model, GPU SKU, or related Azure service that is unavailable in the organization's existing regions or constrained by quota.

DRIVER

AI model, service, SKU, or quota availability.

**FRAMEWORK**

D1 confirms the business requirement. D2 validates data and compliance constraints. D4 validates current model, service, SKU, zone, access, and quota support.

**QUALIFICATION**

Qualified for the defined AI workload requirements; additional dependencies remain conditional until their design is known.

**ARCHETYPE**

Isolated Cloud-Native or AI where self-contained; Connected Cloud-Native or AI where private enterprise dependencies are required.

**CONNECTIVITY PROFILE**

Disconnected Spokes where the workload is self-contained; a connected profile only where actual dependencies justify it.

**PLACEMENT**

Net-new deployment. Revalidate model, SKU, service, regional access, and quota at decision time.

Takeaway. Qualify against the requirements that exist today; do not introduce enterprise connectivity or regional platform capabilities before a workload requirement justifies them.

08 · CUSTOMER SCENARIOS, CONTINUED

3 Acquisition brings an estate in another geography

Portfolio · integration

An acquisition introduces a portfolio of interdependent applications, shared data, and local users in a geography where the organization has little or no existing Azure platform presence.

DRIVER

Business expansion through acquisition.

FRAMEWORK

D1 confirms the business need. D2 evaluates local data obligations. D3 maps portfolio-level dependencies, shared services, and connectivity requirements.

QUALIFICATION

Candidate while material dependencies remain unknown; qualified against defined portfolio requirements once discovery is complete.

ARCHETYPE

Interconnected Application Portfolios.

CONNECTIVITY PROFILE

Remote Hub Connected where cross-region dependencies are acceptable; Minimal Regional Hub where east-west traffic, shared services, latency, or independence justify it.

PLACEMENT

Workloads remain in place initially and relocate selectively as business and integration requirements become known.

🚩 **Takeaway.** Aggregate portfolio dependencies determine the appropriate regional connectivity profile; application count or acquisition status does not.

4 Datacenter exit accelerates migration

Datacenter event · relocation

A colocation contract is ending. Hybrid-connected applications must leave the datacenter, and the organization asks whether the event also requires adoption of another Azure region.

DRIVER

Datacenter and connectivity event.

FRAMEWORK

D3 identifies what must move together. D5 validates hybrid connectivity, datacenter and user geography, reachability, resiliency, and whether existing connectivity suffices.

QUALIFICATION

The existing target region remains qualified for the identified requirements; an additional region remains a candidate for future use.

ARCHETYPE

Hybrid-Connected Applications.

CONNECTIVITY PROFILE

Continue to use an existing strategic geographic hub where it satisfies requirements — no new regional hub solely because another region is available.

PLACEMENT

Most workloads land in the existing qualified region. Revisit the additional option when requirements change.

🚩 **Takeaway.** Remaining in an existing qualified region is a valid outcome. An additional region can remain an option rather than automatically becoming a deployment project.

09

Common misconceptions

PUTTING IT INTO PRACTICE

Multi-region programs often become more complex than necessary when teams start from assumptions instead of workload and platform requirements.

✗ “Every new region needs a Full Regional Hub.”

- ✓ Select the **most efficient profile** that satisfies the identified requirements. Profiles are not maturity stages; any profile can remain the long-term target while it continues to meet them.

✗ “A paired region is our disaster recovery.”

- ✓ Region pairing supports specific Azure platform and service behaviors; it does not automatically provide workload recovery. Recovery must still be designed and validated per workload.

✗ “Qualified means approved for any workload.”

- ✓ Qualification is scoped to defined requirements and evidence at a point in time. Each workload revalidates regional suitability during onboarding.

✗ “A multi-region platform means active-active.”

- ✓ The platform creates regional options. Whether a workload is single-region, primary/recovery, or active across regions is a separate workload-specific resiliency decision.

✗ “Enabling a region means migrating workloads.”

- ✓ Regional qualification and platform readiness are not workload placement. Remaining in the existing region can be the appropriate outcome.

✗ “Another Azure region requires another ExpressRoute circuit.”

- ✓ An additional region does not by itself require another circuit. Add hybrid connectivity where reachability, diversity, failure-domain, latency, routing, or bandwidth requirements justify it.

✗ “Each region needs its own governance model.”

- ✓ Reuse the existing Azure landing-zone governance and operating model; vary regional implementation only where requirements justify it.

✗ “We must choose the perfect regions up front.”

- ✓ Regional requirements and Azure capabilities change. Design the platform so additional qualified options can be introduced without redesigning the broader operating model.



Each correction maps to the framework: profiles (Section 05), zones and pairing (Dimension 4 and Appendix A), qualification and placement (Sections 03 and 06), hybrid connectivity (Dimension 5), and the operating model (Section 07).

09 · PUTTING IT INTO PRACTICE

Getting started

Use this sequence to establish the first additional regional options. The same process scales from one candidate region to a broader portfolio.

01

Name the drivers

Identify what is prompting the discussion — business growth, regulation, service or SKU availability, resiliency, a datacenter or connectivity event, cost, or another measurable requirement.

Drivers

02

Identify workload requirements and archetypes

Start with known workloads and application portfolios. Validate representative dependencies and identify the closest-fitting archetypes. Establish planning guardrails where future requirements are not yet known.

Dimension 3**Archetypes**

03

Shortlist candidate regions

Apply business and geographic requirements with mandatory data and compliance constraints. Remove regions that cannot satisfy mandatory conditions before investing in deeper technical assessment.

Dimensions 1–2

04

Qualify candidates against workload requirements

Validate services, models, SKUs, zone support, dependencies, latency, pricing, regional access, and quota. Record each option as excluded, candidate, conditional, or qualified, with evidence and revalidation triggers.

Dimension 4**Qualification tree**

05

Enable the most efficient platform

Select the connectivity profile, confirm the hybrid connectivity foundation, determine local versus remote shared services, and close the readiness gaps the identified workloads require.

Dimension 5**Profile tree****Readiness checklist**

06

Onboard the first workload

Apply the placement process using current evidence. Revalidate at decision time, govern placement and resiliency decisions, and feed new requirements or platform gaps back into qualification.

Placement flow

WHAT SUCCESS LOOKS LIKE

Being able to qualify a regional option, enable only the platform capabilities it requires, and onboard workloads through a repeatable governed process.

10 CONCLUSION

Design for continuing regional choice

No organization can predict which Azure regions will offer the best mix of services, regulatory fit, business proximity, and workload suitability several years from now.

Design the platform so that perfect prediction is unnecessary.

This framework extends Azure landing-zone principles rather than replacing them. Regional adoption, workload placement, platform expansion, and resiliency can evolve without redesigning the broader platform for every new region.

Workload-led placement



Qualified regional options, Application Landing Zone archetypes, and current workload requirements allow applications to use the regional and platform capabilities appropriate to their needs.

Adaptable regional platform



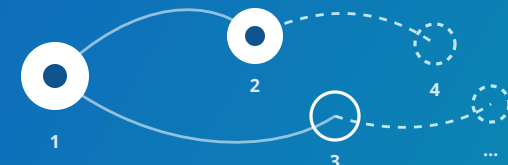
Strategic geographic hubs, resilient hybrid connectivity, and regional connectivity profiles distribute capabilities according to workload requirements — without an identical footprint in every region.

Consistent operating model



Keep the landing-zone guardrails and operating standards stable across the Azure estate.

The goal is not to select the perfect two regions. **It is to make additional regions significantly easier to adopt when business and workload requirements justify them.**





APPENDIX A · REGIONAL DEPENDENCIES AND RESILIENCY

Paired vs. nonpaired regions

Region pairing is one input into resiliency design, not the definition of it. Azure supports resilient architectures using paired regions, nonpaired regions, or both.



Azure paired regions

PLATFORM-ASSISTED REGIONAL RELATIONSHIP

- Microsoft defines region pairs, usually within the same Azure geography, although asymmetric and cross-geography exceptions exist.
- Some Azure services use the paired region for built-in geo-replication or geo-redundancy — for example, geo-redundant storage, Key Vault, and Azure Backup Cross Region Restore.
- Azure uses region pairs for sequential platform updates and recovery prioritization.
- Deploying into both members of a pair does **not** automatically provide workload high availability, disaster recovery, or failover.

Consider: built-in service behavior can constrain the secondary region to the Azure-defined pair. Some paired regions have restricted access (for example, Brazil Southeast, Germany North, Norway West, Switzerland West), and services can have different regional capabilities.



Region of choice

NONPAIRED · ARCHITECTURE-LED RECOVERY

- Many newer Azure regions are nonpaired and use availability zones as an important part of their intra-region resiliency model.
- A multi-region workload can use a nonpaired or customer-selected secondary region where the required Azure services support that regional combination — through Azure service-based or ISV-based disaster recovery.
- Provides flexibility for residency, sovereignty, latency, service availability, or business-continuity requirements.
- The recovery architecture, replication method, dependencies, and failover process must be explicitly designed and validated per workload.

Pairing can provide useful platform and service capabilities; it does not replace a tested workload recovery design.

“Deploying resources to a region in a pair doesn’t automatically make them more resilient, nor does it provide automatic high availability, disaster recovery capabilities, or failover.”

MICROSOFT LEARN · AZURE REGION PAIRS AND NONPAIRED REGIONS

RELATIONSHIP TO DIMENSION 4

Assess availability zones, region pairing, and service-specific cross-region behavior separately (Figure 11). Whether a regional combination works depends on the workload’s requirements and the capabilities of each service it uses.

APPENDIX A · SERVICE-LEVEL DETAIL

Three services that need special attention

These three services illustrate why regional qualification cannot stop at “the service exists in both regions.” Cross-region behavior can differ significantly by service and regional combination.

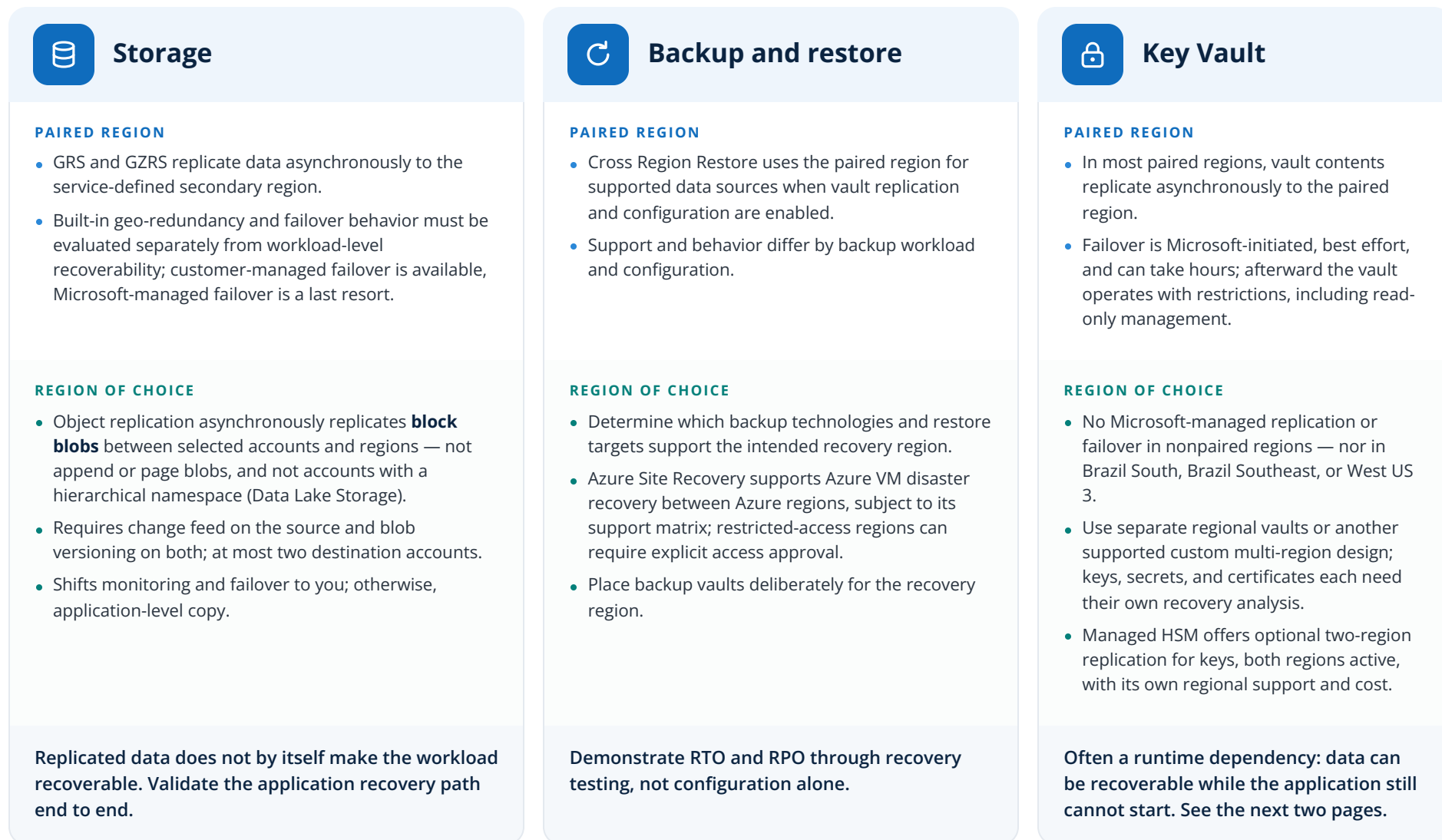


FIGURE 26 Service behavior differs by regional combination. Built-in replication for all three services targets the paired region; region-of-choice designs need explicit replication, restore, and secret-management patterns. Confirm current capabilities on Microsoft Learn at decision time.

APPENDIX A · KEY VAULT

Key Vault is a workload-specific dependency



*“Data can be recoverable while the application **still cannot start.**”*



A runtime dependency

Applications can require keys, secrets, certificates, DNS resolution, and private connectivity simply to initialize. Recovered data is insufficient if runtime dependencies are unavailable in the recovery region.



Keys, secrets, and certificates are not interchangeable

Each needs its own recovery analysis. Secrets and certificates can often be provisioned into regional vaults through controlled deployment processes; customer-managed keys have additional constraints.



Managed failover may not meet the objective

Microsoft controls paired-region failover and describes it as best effort. A prolonged outage can mean hours before failover, and the secondary operates with restrictions (read-only).



Managed HSM addresses a different requirement

Optional two-region replication of key material, roles, and permissions, with both regions serving requests — for **keys**, not secrets or certificates, at a higher cost profile. Not every region can be an extended region.

NOTE Microsoft documents custom multi-region Key Vault approaches for nonpaired regions, paired regions without Microsoft-managed replication, and workloads whose recovery objectives cannot be met by built-in failover.

APPENDIX A · THE RECOMMENDED PATTERN

Independent Key Vault per workload per region

For controlled region-of-choice recovery, give each active or recovery region the Key Vault capability it needs **before failover** is required.

- ✓ **Independent regional vault** — each workload uses the vault in its local region rather than depending at runtime on a vault in another region.
- ✓ **Deploy configuration consistently** — infrastructure as code provisions vault configuration, access, private connectivity, monitoring, and policy. An approved secret-management process populates secrets and certificates; do not treat the IaC repository itself as a store for secret values.
- ✓ **Do not make failover the synchronization strategy** — populate and validate runtime dependencies before an outage, not by creating or restoring a vault after the primary region has failed.
- ✓ **Design cryptographic keys separately** — customer-managed keys cannot always be reproduced like secrets. Choose a key-resiliency pattern: Key Vault backup/restore within its constraints, Managed HSM multi-region, or a workload-specific design.

DR in practice: when the workload activates in the recovery region, the required vault and runtime dependencies are already available and validated there.

Microsoft Learn also documents backup and restore between vaults; backups can only be restored within the same subscription and Azure geography. Learn: [Custom multi-region solutions for resiliency](#).

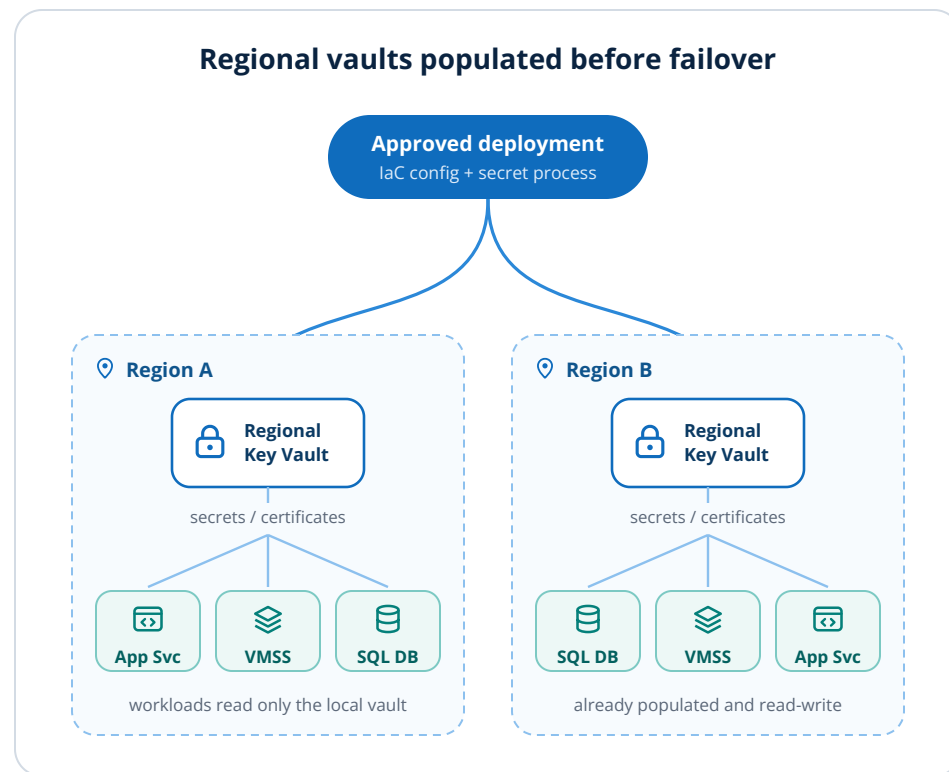


FIGURE 27 Pre-provisioned regional vaults. Applies to secrets and certificates. Design customer-managed keys (for example, SQL TDE) separately — see Scenario R3.

APPENDIX A · RESILIENCY SCENARIOS

Three resiliency scenarios

These examples show how service-specific regional behavior changes workload recovery design. They complement the broader placement scenarios in Section 08.

SCENARIO R1

A workload in a nonpaired region requires DR

A workload operates in a zone-enabled region without a pair and must recover in another Azure region.

STORAGE	A supported region-of-choice pattern such as object replication where suitable, or application-level replication where required (for example, Data Lake Storage).
COMPUTE	Azure Site Recovery for applicable VMs, or redeploy stateless compute from infrastructure as code.
KEY VAULT	Pre-provision the regional vault; runtime secrets, certificates, and any key strategy available before recovery.
BACKUP	Place backup and restore capabilities deliberately for the intended recovery region.

Takeaway. A nonpaired region can participate in a resilient multi-region architecture, but each workload layer needs an explicit recovery design.

SCENARIO R2

"We're paired, so we have DR"

Data is geo-replicated to the paired region, but the complete application has never been failed over and tested.

SYMPTOM	Data is recoverable, but the application cannot operate in the secondary region.
POSSIBLE CAUSES	Key Vault or another runtime dependency unavailable (failover not yet occurred, or read-only); DNS or private-endpoint configuration incomplete; compute or shared services not recoverable; dependencies missing from the recovery design.
CORRECTION	Treat data, identity, secrets and keys, DNS, networking, shared services, and compute as workload dependencies. Validate end-to-end recovery against RTO and RPO.

Takeaway. Data recovery is not application recovery, and region pairing does not replace a tested workload recovery plan.

SCENARIO R3

Customer-managed keys across two regions

A workload uses customer-managed encryption keys (for example, database TDE) and must operate or recover across two regions.

CONSTRAINT	Key material has different recovery and replication characteristics from secrets and certificates; it cannot be assumed to be recreated by the pipeline.
OPTIONS	Key Vault behavior for the selected regions (paired-region replication where available); backup and restore within the same subscription and geography; Azure Managed HSM multi-region replication; or another workload-specific design.
VALIDATE	Key availability, access, private connectivity, permissions, and application behavior in the secondary region before the recovery path is considered complete.

Takeaway. Design key availability explicitly. For some workloads, key management determines which recovery regions are viable.

FIGURE 28 Resiliency scenarios. Representative situations, not drawn from a specific customer. Confirm current service capabilities on Microsoft Learn at decision time.

REFERENCES AND FURTHER READING

Microsoft Learn and Azure resources cited in this paper

MULTI-REGION WORKLOADS

Using availability zones and regions — Azure Well-Architected Framework

Deployment approaches, including multi-region deployments. learn.microsoft.com/azure/well-architected/design-guides/regions-availability-zones

DIMENSION 1 · BUSINESS AND GEOGRAPHY

Select Azure regions

Aligning regional choices with user locations, business requirements, and deployment considerations. learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-setup-guide/regions

Azure geographies

Azure's geographic footprint and locations relevant to markets and operations. azure.microsoft.com/explore/global-infrastructure/geographies

DIMENSION 2 · DATA AND COMPLIANCE

Data residency in Azure

Customer-data location, geographic boundaries, and service-specific exceptions. azure.microsoft.com/explore/global-infrastructure/data-residency

Azure compliance documentation

Regional, industry-specific, and global compliance references. learn.microsoft.com/azure/compliance

Reliability and sovereignty in Azure

Sovereignty and regional selection, replication destinations, encryption, and operational control. learn.microsoft.com/azure/reliability/concept-reliability-sovereignty

DIMENSION 3 · WORKLOAD ARCHETYPES AND DEPENDENCIES

Architecture styles

N-tier, microservices, event-driven, and other styles. learn.microsoft.com/azure/architecture/guide/architecture-styles

Evaluate a cloud workload for relocation

Dependency discovery, workload ownership, acceptable disruption, and target-region supportability. learn.microsoft.com/azure/azure-resource-manager/management/relocate-evaluate

Relocate cloud workloads

The Azure-to-Azure relocation process; assessment distinct from migration and cutover. learn.microsoft.com/azure/azure-resource-manager/management/relocate-index

DIMENSION 4 · REGIONAL CAPABILITY AND AVAILABILITY

Azure region pairs and nonpaired regions

learn.microsoft.com/azure/reliability/regions-paired

Azure services that support multiple regions

learn.microsoft.com/azure/reliability/regions-multiregion-support

Azure services that support availability zones

learn.microsoft.com/azure/reliability/availability-zones-service-support

Zonal resources and zone resiliency

learn.microsoft.com/azure/reliability/availability-zones-zonal-resource-resiliency

Enable zone resiliency for Azure workloads

learn.microsoft.com/azure/reliability/availability-zones-enable-zone-resiliency

DIMENSION 5 · PLATFORM ENABLEMENT AND CONNECTIVITY

Azure landing zone design areas and conceptual architecture

Identity, security, governance, management, connectivity, and deployment automation. learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-areas

Designing for disaster recovery with ExpressRoute private peering

Peering-location diversity, redundant paths, and failover. learn.microsoft.com/azure/expressroute/designing-for-disaster-recovery-with-expressroute-privatepeering

Network topology and connectivity

Connectivity between workloads, shared services, and the broader platform. learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/network-topology-and-connectivity

Define an Azure network topology

Topology approaches based on cross-region connectivity and routing requirements. learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/define-an-azure-network-topology

REFERENCES AND FURTHER READING · CONTINUED

APPENDIX A · REGIONAL DEPENDENCIES AND RESILIENCY

Azure services that support multiple regions

Built-in paired-region replication vs. region-of-choice options by service.

learn.microsoft.com/azure/reliability/regions-multiregion-support

Object replication for block blobs

Supported blob types, prerequisites, and limits for region-of-choice replication.

learn.microsoft.com/azure/storage/blobs/object-replication-overview

Azure Storage redundancy

LRS, ZRS, GRS, and GZRS options and failover.

learn.microsoft.com/azure/storage/common/storage-redundancy

Restore Azure VMs — Cross Region Restore

Restoring VMs to the paired secondary region.

learn.microsoft.com/azure/backup/backup-azure-arm-restore-vm

Azure Site Recovery support matrix

Region support for Azure-to-Azure disaster recovery, including restricted regions.

learn.microsoft.com/azure/site-recovery/azure-to-azure-support-matrix

Azure Key Vault availability and redundancy

Replication, Microsoft-managed failover, and backup and restore limits.

learn.microsoft.com/azure/key-vault/general/disaster-recovery-guidance

Reliability in Azure Key Vault

Custom multi-region solutions for resiliency.

learn.microsoft.com/azure/reliability/reliability-key-vault

Managed HSM multi-region replication

Extending a Managed HSM pool to an additional region.

learn.microsoft.com/azure/key-vault/managed-hsm/multi-region-replication

A NOTE ON THE FIGURES

Region names, service placements, and estates shown in the figures are illustrative. They explain the framework; they do not recommend specific Azure regions or topologies.

SERVICE CAPABILITIES CHANGE

Regional service behavior described in Appendix A reflects Microsoft Learn at the time of writing. Confirm current capabilities at decision time.

The adaptable **multi-region** Azure platform

Extending Azure landing zones for adaptable regional placement through the Regional Qualification and Enablement Framework, Application Landing Zone Archetypes, and Regional Platform Connectivity Profiles.

Multi-Region Platform Whitepaper · September 2026

